AIUB Sport Portal Management System

Course Name

ADVANCE DATABASE MANAGEMENT SYSTEM

Section: C

Group Members

Name	ID	Contribution
MD. Nashib Khan (Project Lead)	22-49873-3	Introduction, Project proposal ,Scenario description, User Interface design, PL/SQL, Database design (25%)
MD Momen Sha	24-56434-1	Idea Generation, User Interface, Advance PL/SQL, Database design, Schema Diagram , Coding(frontend &backend) (25%)
MD. Al Amin	22-48081-2	ER Diagram, Normalization, Schema Diagram, Basic PL/SQL, Advance PL/SQL (25%)
MD. Sharif Iqbal	22-49924-3	Project proposal, Table Creation using SQL, Basic PL/SQL , Coding(frontend), Schema Diagram (25%)
Pranto Sen	23-52029-2	Introduction, Coding(Backend) ,Data Insertion Using SQL, Schema Diagram ,Conclusion . (25%)

Contents

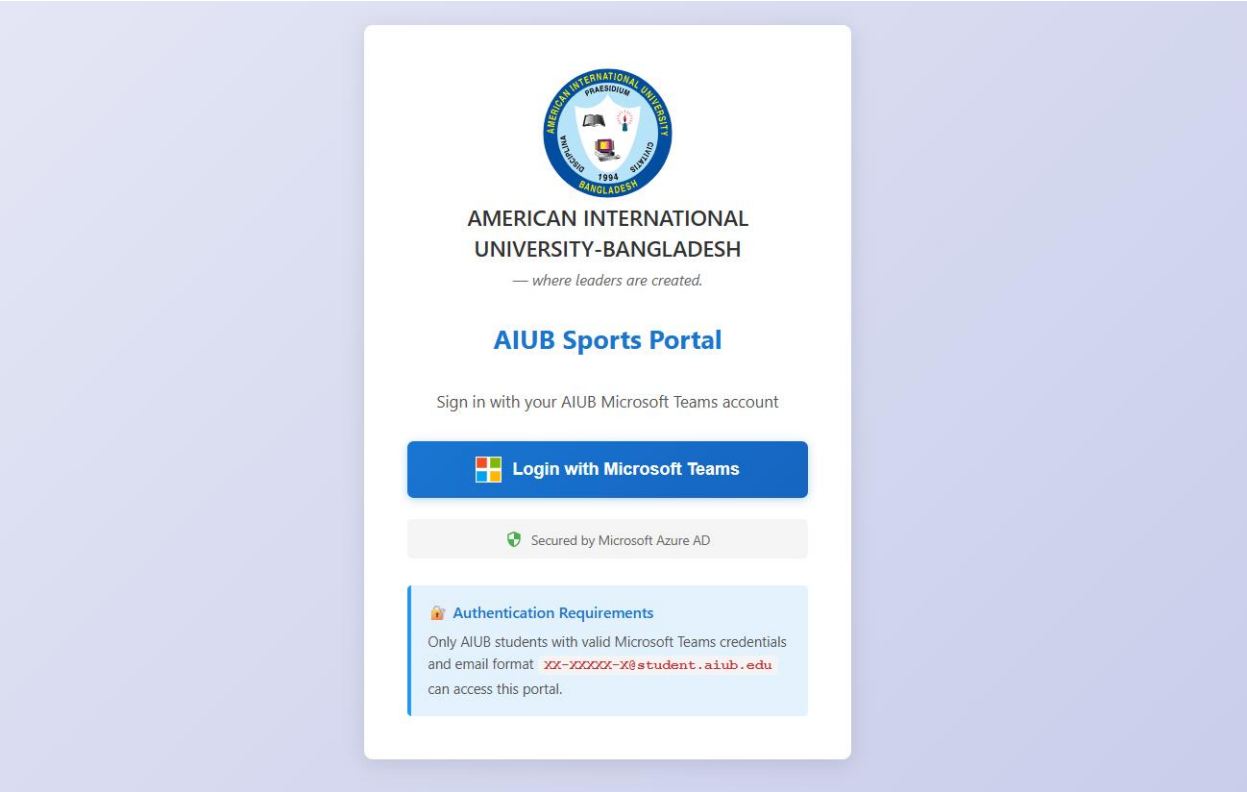
Introduction:

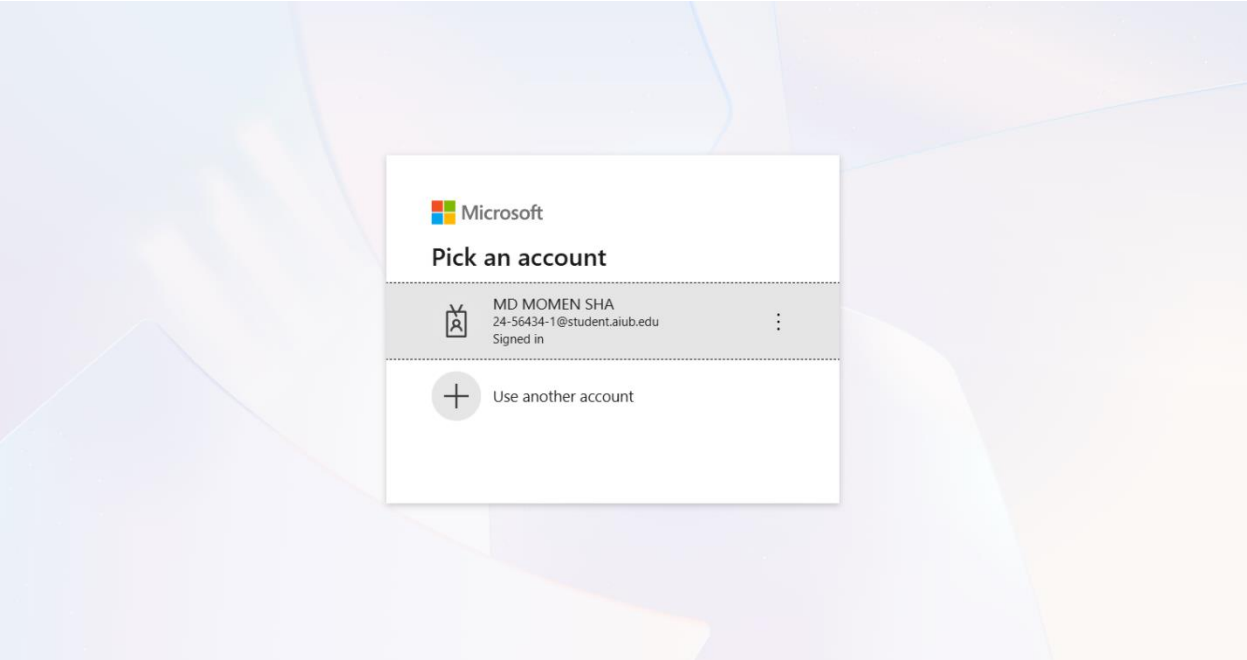
The AIUB Sports Portal is a web-based platform designed to make sports activities at American International University–Bangladesh easier, faster, and more organized. Every semester, many students participate in tournaments, but registration is usually manual, time-consuming, and difficult to manage. This project solves that problem by creating a digital system where students can log in with their official AIUB accounts, complete their profiles, view ongoing tournaments, and register for games. Admins can also create tournaments, manage participants, and handle all sports-related tasks from one place. In simple terms, the portal brings the entire sports management process online so both students and AIUB authorities can save time and reduce mistakes.

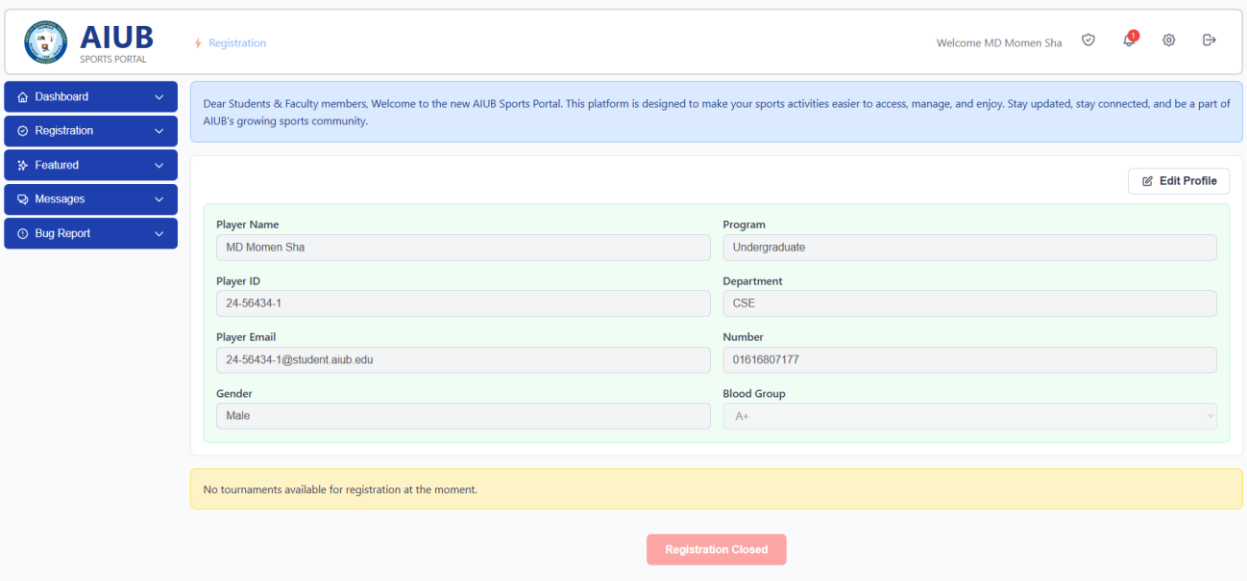
Project Proposal:

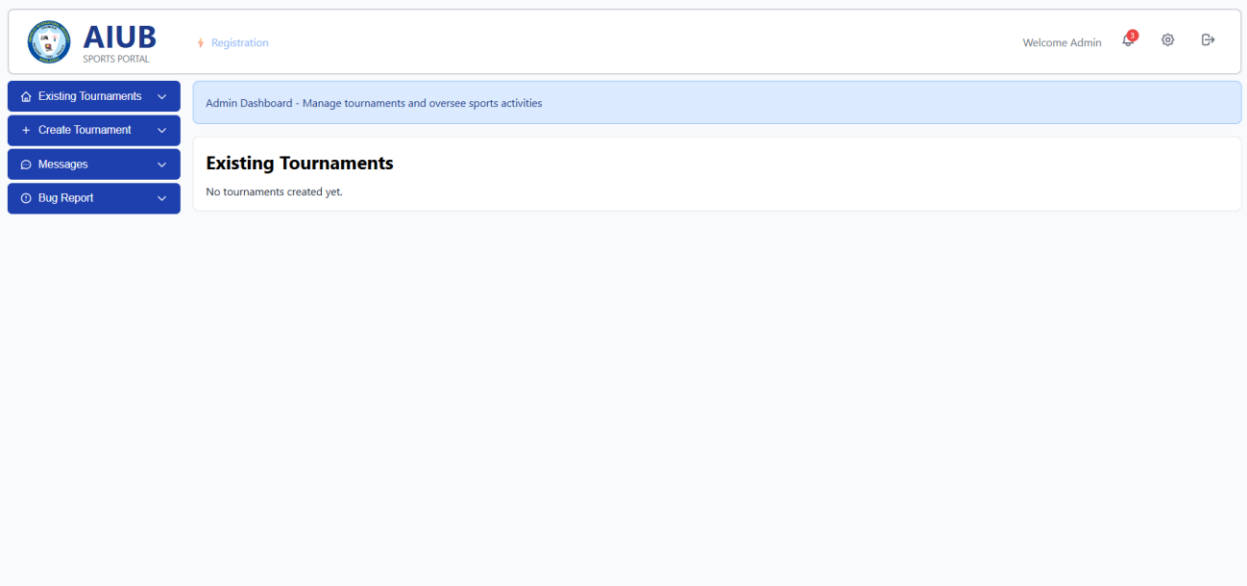
This project proposes the development of a complete sports management web portal for AIUB that integrates authentication, tournament management, and game registration within a secure and intuitive interface. The goal is to eliminate the traditional paperwork-based registration system and replace it with a reliable digital system that ensures accuracy, safety, and real-time data handling. The portal will allow students to log in using Microsoft Azure Active Directory, which guarantees identity verification and restricts access only to authentic AIUB students. Once logged in, students can complete their profiles, view ongoing tournaments, and register for specific sports categories. Administrators will be provided with a dedicated dashboard where they can create tournaments, upload tournament images, set registration deadlines, add game categories, and manage participants. To maintain fairness and data accuracy, the system includes several important business rules, such as limiting the number of name edits, locking sensitive profile fields after initial setup, and enforcing registration deadlines. The proposal supports scalable architecture built with Node.js, Express.js, Oracle Database (10g), and pure HTML/CSS/JavaScript on the frontend. The platform ensures secure communication, strict input validation, and structured database design. By integrating PL/SQL stored procedures, triggers, and sequences, the system achieves high data integrity and reliable automation. Overall, the AIUB Sports Portal aims to provide AIUB students and admins with professional, automated, transparent, and trustworthy sports management experience.

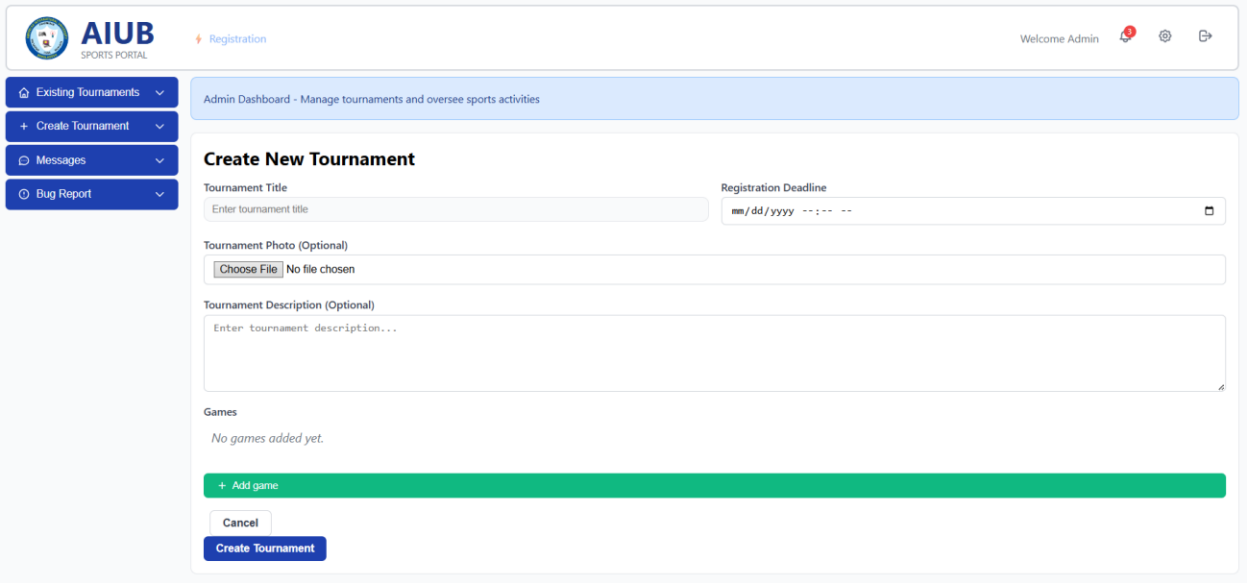
User Interface Planning:

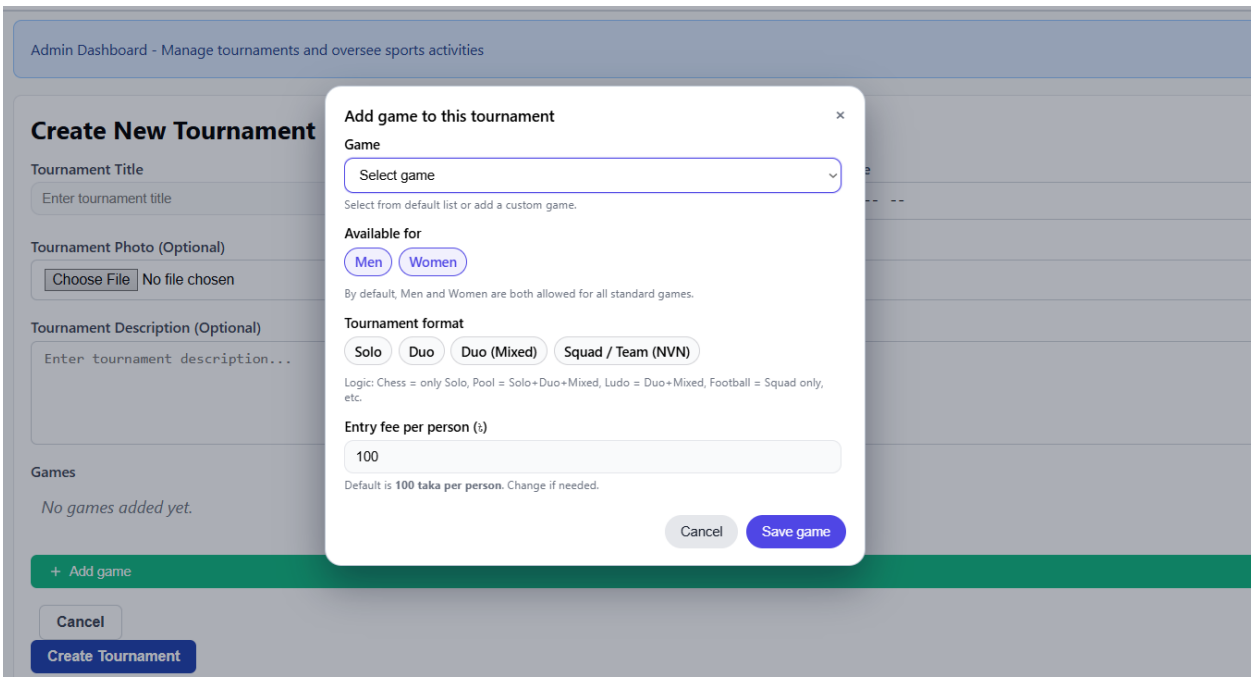





AIUB
SPORTS PORTAL
Registration
Welcome MD Momen Sha
Dashboard
Registration
Featured
Messages
Bug Report
Dear Students & Faculty members, Welcome to the new AIUB Sports Portal. This platform is designed to make your sports activities easier to access, manage, and enjoy. Stay updated, stay connected, and be a part of AIUB's growing sports community.
Edit Profile
Player Name
MD Momen Sha
Program
Undergraduate
Player ID
24-56434-1
Department
CSE
Player Email
24-56434-1@student.aiub.edu
Number
01616807177
Gender
Male
Blood Group
A+
No tournaments available for registration at the moment.
Registration Closed

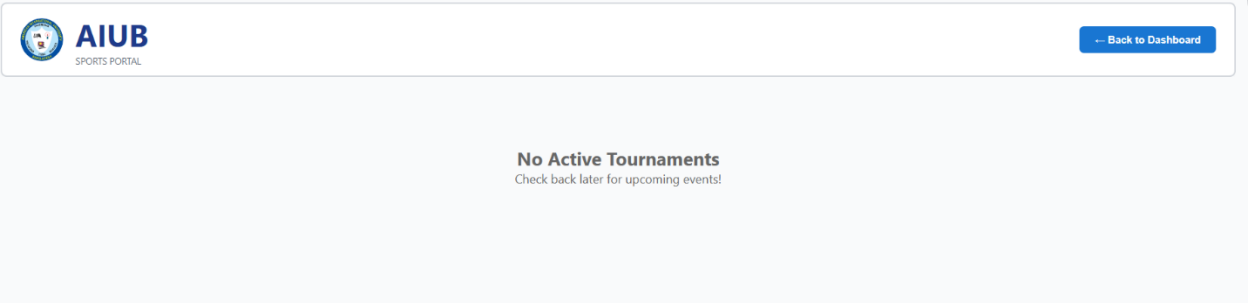

AIUB
SPORTS PORTAL
Registration
Welcome Admin
Existing Tournaments
Create Tournament
Messages
Bug Report
Admin Dashboard - Manage tournaments and oversee sports activities
Existing Tournaments
No tournaments created yet.


AIUB
SPORTS PORTAL
Registration
Welcome Admin
Existing Tournaments
Create Tournament
Messages
Bug Report
Admin Dashboard - Manage tournaments and oversee sports activities
Create New Tournament
Tournament Title
Enter tournament title
Registration Deadline
mm/dd/yyyy --:-- --
Tournament Photo (Optional)
Choose File No file chosen
Tournament Description (Optional)
Enter tournament description...
Games
No games added yet.
+ Add game
Cancel
Create Tournament


Admin Dashboard - Manage tournaments and oversee sports activities
Create New Tournament
Tournament Title
Enter tournament title
Tournament Photo (Optional)
Choose File No file chosen
Tournament Description (Optional)
Enter tournament description...
Games
No games added yet.
+ Add game
Cancel
Create Tournament
Add game to this tournament
Game
Select game
Select from default list or add a custom game.
Available for
Men
Women
By default, Men and Women are both allowed for all standard games.
Tournament format
Solo
Duo
Duo (Mixed)
Squad / Team (NVN)
Logic: Chess = only Solo, Pool = Solo+Duo+Mixed, Ludo = Duo+Mixed, Football = Squad only, etc.
Entry fee per person (৳)
100
Default is 100 taka per person. Change if needed.
Cancel
Save game


Games
Badminton (Male)
Type: Solo • Fee: 100 BDT
Delete
Badminton (Female)
Type: Solo • Fee: 100 BDT
Delete
Badminton (Male)
Type: Duo • Fee: 100 BDT
Delete
Badminton (Female)
Type: Duo • Fee: 100 BDT
Delete
Badminton (Mix)
Type: Duo (Mixed) • Fee: 100 BDT
Delete
+ Add game
Cancel
Create Tournament

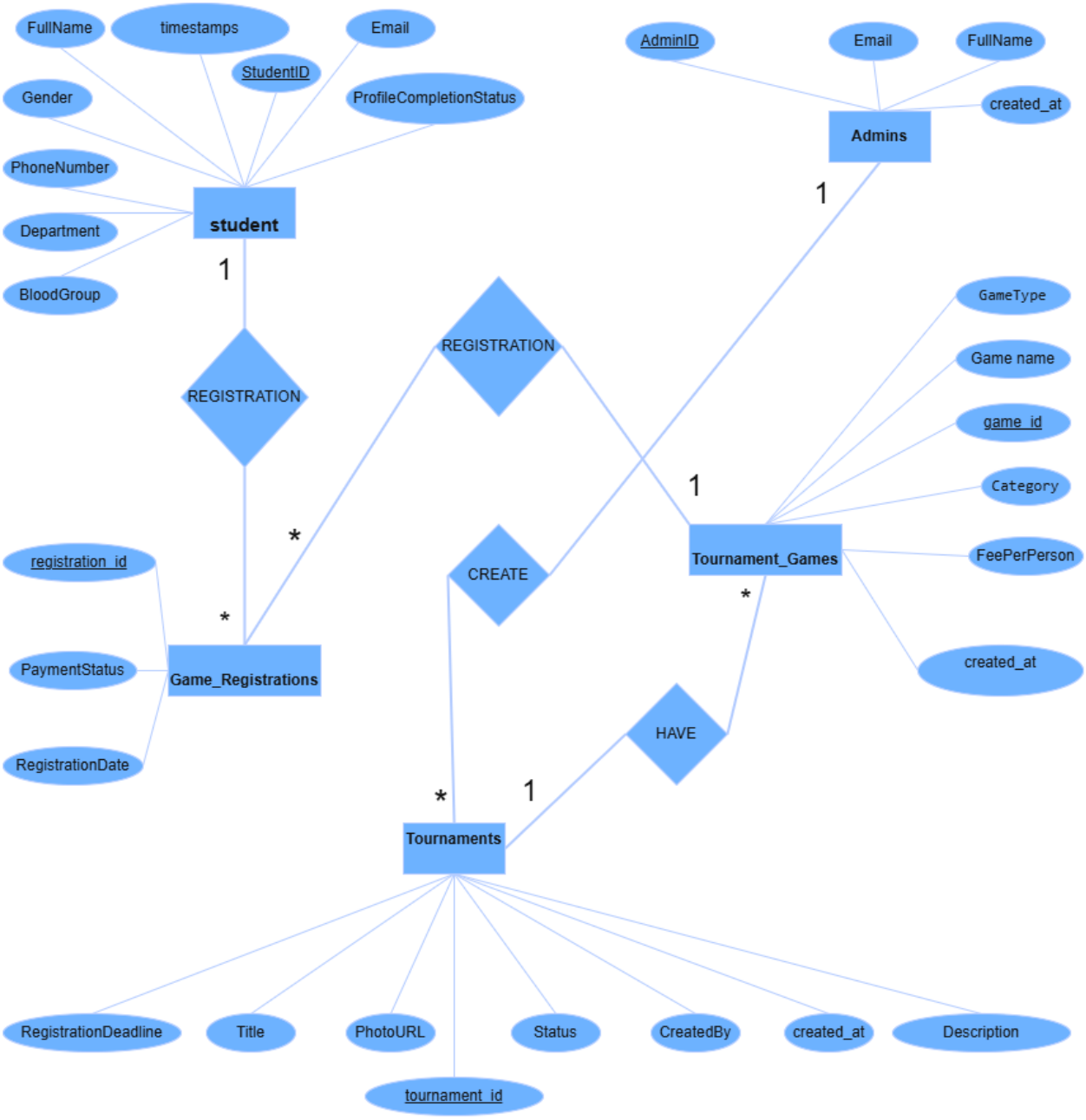

Games
Showing 5 existing games. Add more or update below.
Male Category Games
Badminton (Duo)
100 BDT
Badminton (Solo)
100 BDT
Female Category Games
Badminton (Duo)
100 BDT
Badminton (Solo)
100 BDT
Mix Category Games
Badminton (Custom)
100 BDT
+ Add game
Cancel
Update Tournament


AIUB
SPORTS PORTAL
← Back to Dashboard
No Active Tournaments
Check back later for upcoming events!

Scenario Description

In an AIUB Sports Management System, students can register for different games offered within tournaments. A game must have at least one student registered, although students are not required to join any game. Each game belongs to a tournament, and it cannot exist without one. Tournaments are created and managed by admins. Students and admins are stored separately in the system. Students log in using Microsoft Teams and maintain a profile containing details like name, gender, department, program level, phone number, and blood group. Some information, such as gender and department, becomes locked after the first submission, while the name can only be edited a limited number of times. Each tournament includes a title, description, registration deadline, and an image. An admin can create multiple tournaments, and every tournament can have multiple games such as football, cricket, or badminton, where each game is classified under male, female, or mixed categories. Students can register for any available game as long as the tournament registration deadline has not passed. They may join multiple games but cannot register more than once for the same game. Each registration records the registration date and payment status, which may be pending, paid, or failed. Unique IDs identify all students, admins, tournaments, games, and registrations. Foreign keys ensure proper relationships among these entities. To keep data consistent, triggers and sequences automatically generate IDs and timestamps. Overall, the system represents a real-world sports portal where students take part in tournaments and games, while admins handle tournament creation and game management.

ER Diagram


FullName
timestamps
Email
StudentID
Gender
ProfileCompletionStatus
PhoneNumber
student
Department
BloodGroup
1
REGISTRATION
*
registration_id
*
PaymentStatus
Game_Registrations
RegistrationDate
REGISTRATION
CREATE
AdminID
Email
FullName
created_at
Admins
1
1
GameType
Game name
game_id
Category
Tournament_Games
FeePerPerson
*
created_at
HAVE
*
1
Tournaments
RegistrationDeadline
Title
PhotoURL
Status
CreatedBy
created_at
Description
tournament_id

Normalization

RELATION–1 : USER REGISTRATION: (User ↔ Game)

UNF:

Student_registration(student_id_, email, full_name, gender, program_level, department, blood_group, registration_id, registration_date, payment_status)

1NF:

There is no multivalued attribute. All attributes are atomic. Relation already in 1NF.

2NF:

1. student(_student_id_, email, full_name, gender, program_level, department, blood_group)

2. GAME_REGISTRATIONS(_registration _id, registration_date, payment_status)

Partial Dependency:

student_id, email, full_name, gender, program_level, department, blood_group
(depend only on student_id), registration_date, payment_status (depend on registration_id)

3NF:

No transitive dependency Relation already in 3NF.

Table Creation:

1. Student(**student_id**_, email, full_name, gender, program_level, department, blood_group)

2. GAME_REGISTRATIONS(_**registration _id**, registration_date, payment_status)

RELATION–2 : TOURNAMENT_CREATION

UNF:

tournament_creation(a_id, admin_name, admin_email,created_at tournament_id, title, photo_url, registration_deadline, status, description, created_at,CreatedBy)

1NF:

There is no multivalued attribute. Relation already in 1NF

2NF:

1. ADMINS (a_id, admin_name, admin_email,created_at)

2. TOURNAMENTS(t_id, title, photo_url, registration_deadline, status, description, created_at,CreatedBy)

Partial Dependency:

admin_name, admin_email (depend on a_id) , registration_deadline, status (depend on t_id)

3NF:

No transitive dependency Relation already in 3NF.

Table Creation:

1. ADMINS(**a_id**, admin_name, admin_email,created_at)

2. TOURNAMENTS(**t_id**, title, photo_url,registration_deadline, status,description, created_at,CreatedBy)

RELATION–3 : TOURNAMENT_have_Tournament_Games:

UNF:

TOURNAMENTS_HAVE_GAMES(t_id, title, photo_url, registration_deadline, status, created_by, description, g_id, category, game_name, game_type, fee_per_person,created_at)

1NF:

There is no multivalued attribute.All attributes are atomic. Relation already in 1NF

2NF:

1. TOURNAMENTS(t_id (PK),title, photo_url, registration_deadline,status, created_by, description)
2. TOURNAMENT_GAMES(g_id (PK),t_id (FK) ,category, game_name, game_type, fee_per_person)

Composite key: (t_id, g_id)

Partial dependencies:

title, photo_url, registration_deadline, status, created_by, description depends on t_id

category, game_name, game_type, fee_per_person depends on g_id

3NF:

No transitive dependency Relation already in 3NF.

Table Creation:

1. TOURNAMENTS (**t_id**, title, photo_url, registration_deadline, status, description, created_at,CreatedBy)
2. TOURNAMENT_GAMES(**g_id** ,category,game_name,game_type, fee_per_person, created_at)

RELATION–4 : Game_Registrations ↔ Tournament_Games

UNF:

GAME_REGISTRATION (registration_id, registration_deadline, PaymentStatus g_id , category, game_name, game_type, fee_per_person, created_at)

1NF:

There is no multivalued attribute. All attributes are atomic. Relation already in 1NF.

2NF:

1. Game_Registrations (registration_id, PaymentStatus, RegistrationDate)

2. TOURNAMENT_GAMES(g_id , category,game_name,game_type, fee_per_person)

Composite PK = (reg_id, g_id)

Partial dependencies: registration_deadline, PaymentStatus (reg_id)

category, game_name, game_type, fee_per_person, created_at (g_id)

3NF:

No transitive dependency Relation already in 3NF.

Table Creation:

1. Game_Registrations (**registration_id**, PaymentStatus, RegistrationDate)

2. TOURNAMENT_GAMES**(g_id** , category, game_name, game_type, fee_per_person)

Temporary Tables:

1.**student_id**, email, full_name, gender, program_level, department, blood_group

~~2.registration_id, registration_date, payment_status, student_id (FK)~~

3. **a_id**, admin_name, admin_email, created_at

~~4. t_id, title, photo_url, registration_deadline, status, description, created_at, CreatedBy, a_id(FK)~~

5. **t_id**, title, photo_url, registration_deadline, status, description, created_at, CreatedBy, a_id(FK)

~~6.g_id, category, game_name, game_type, fee_per_person, created_at, t_id(FK)~~

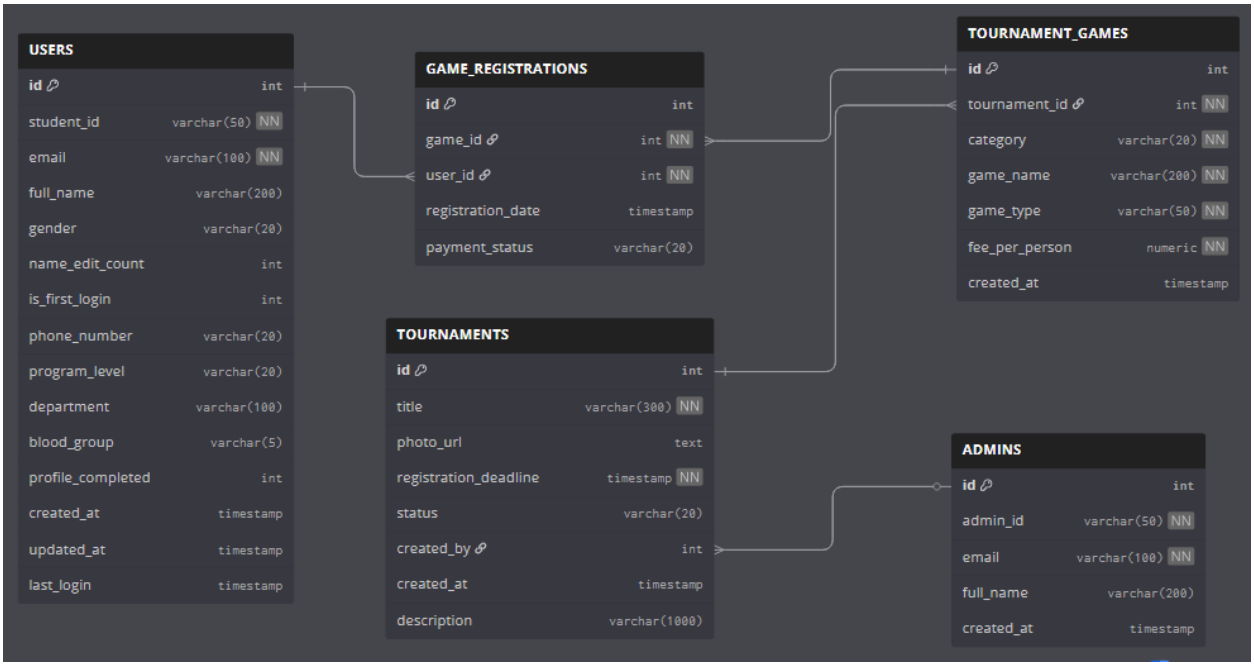
7.**registration_id**, registration_date, payment_status, student_id (FK), t_id(FK)

8. **g_id**, category, game_name, game_type, fee_per_person, created_at, t_id(FK)

Final Tables:

1. **student_id**, email, full_name, gender, program_level, department, blood_group
2. **a_id**, admin_name, admin_email, created_at
3. **t_id**, title, photo_url, registration_deadline, status, description, created_at, CreatedBy, a_id(FK)
4. **registration_id**, registration_date, payment_status, student_id (FK), t_id(FK)
5. **g_id**, category, game_name, game_type, fee_per_person, created_at, t_id(FK)

Schema Diagram



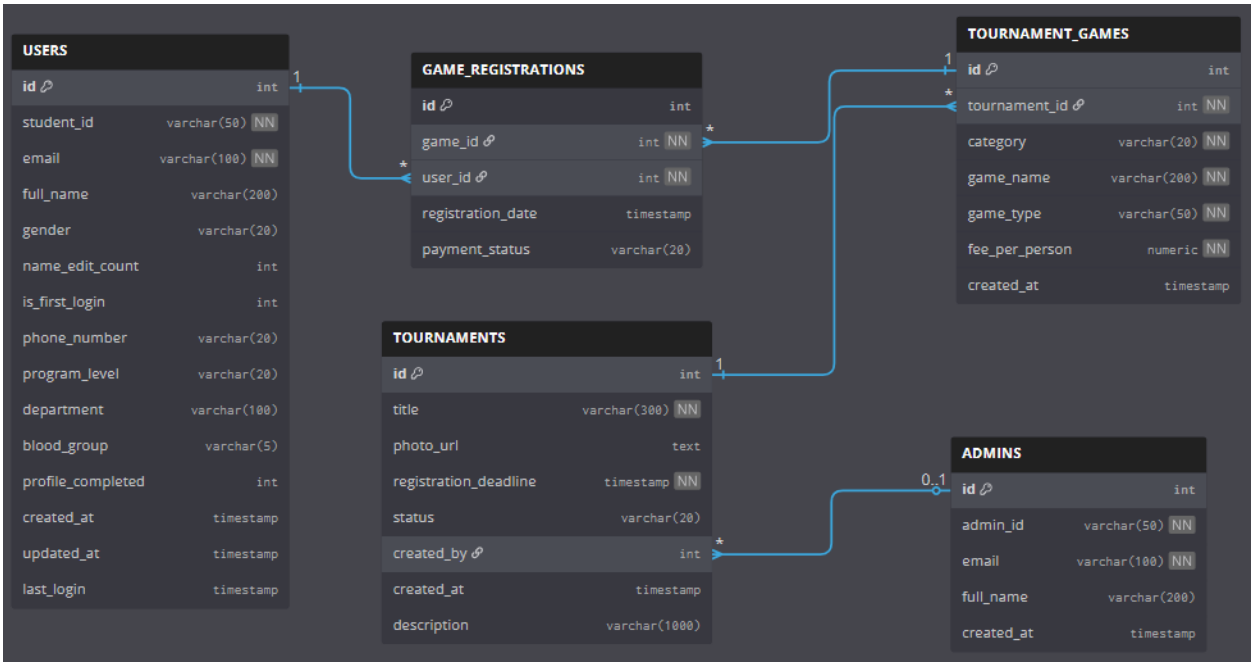
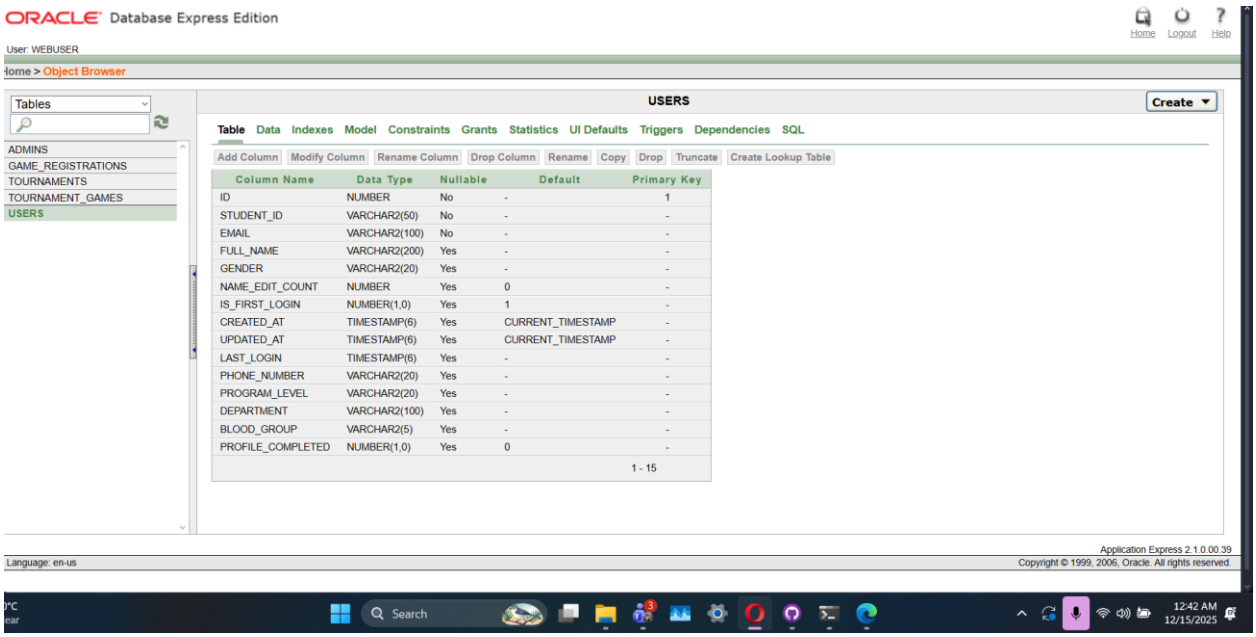


Table creation using SQL

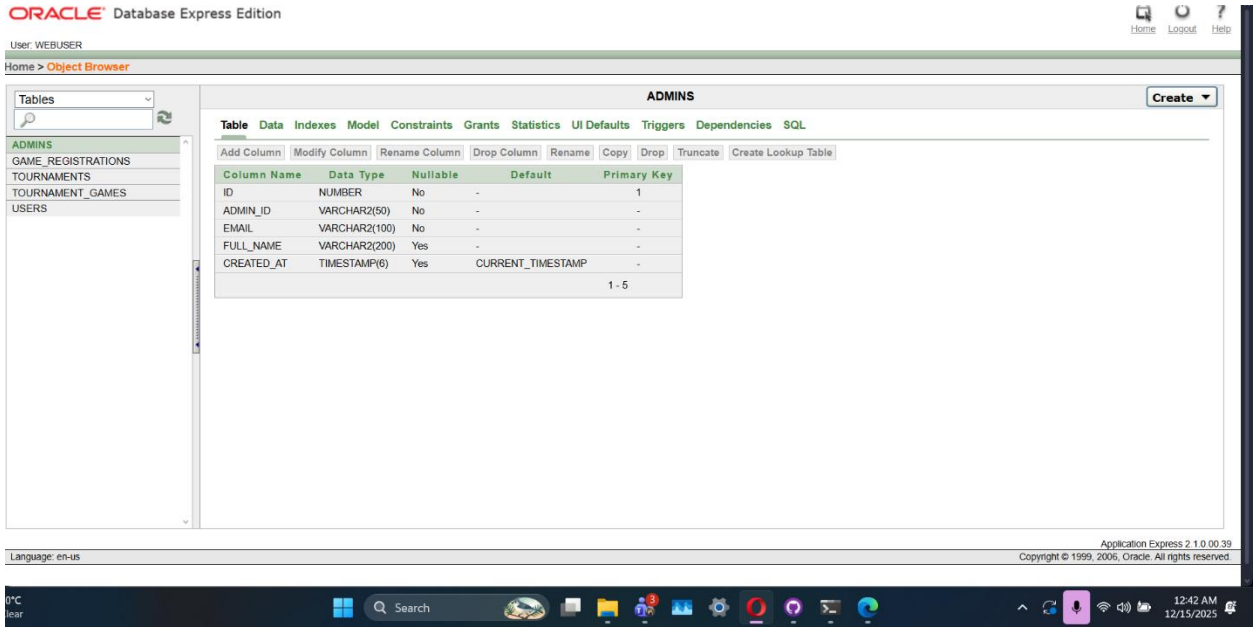
Users Table

```
CREATE TABLE users (
    id NUMBER PRIMARY KEY,
    student_id VARCHAR2(50) NOT NULL UNIQUE,
    email VARCHAR2(100) NOT NULL UNIQUE,
    full_name VARCHAR2(200),
    gender VARCHAR2(20),
    name_edit_count NUMBER DEFAULT 0,
    is_first_login NUMBER(1) DEFAULT 1,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_login TIMESTAMP,
    phone_number VARCHAR2(20),
    program_level VARCHAR2(20),
    department VARCHAR2(100),
    blood_group VARCHAR2(5),
    profile_completed NUMBER(1) DEFAULT 0,
    CONSTRAINT chk_gender CHECK (gender IN ('Male', 'Female', 'Other', NULL)),
    CONSTRAINT chk_edit_count CHECK (name_edit_count >= 0 AND name_edit_count <=
3),
    CONSTRAINT chk_first_login CHECK (is_first_login IN (0, 1)),
    CONSTRAINT chk_program_level CHECK (program_level IN ('Undergraduate',
'Postgraduate', NULL)),
    CONSTRAINT chk_blood_group CHECK (blood_group IN ('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-
', 'O+', 'O-', NULL))
);
```



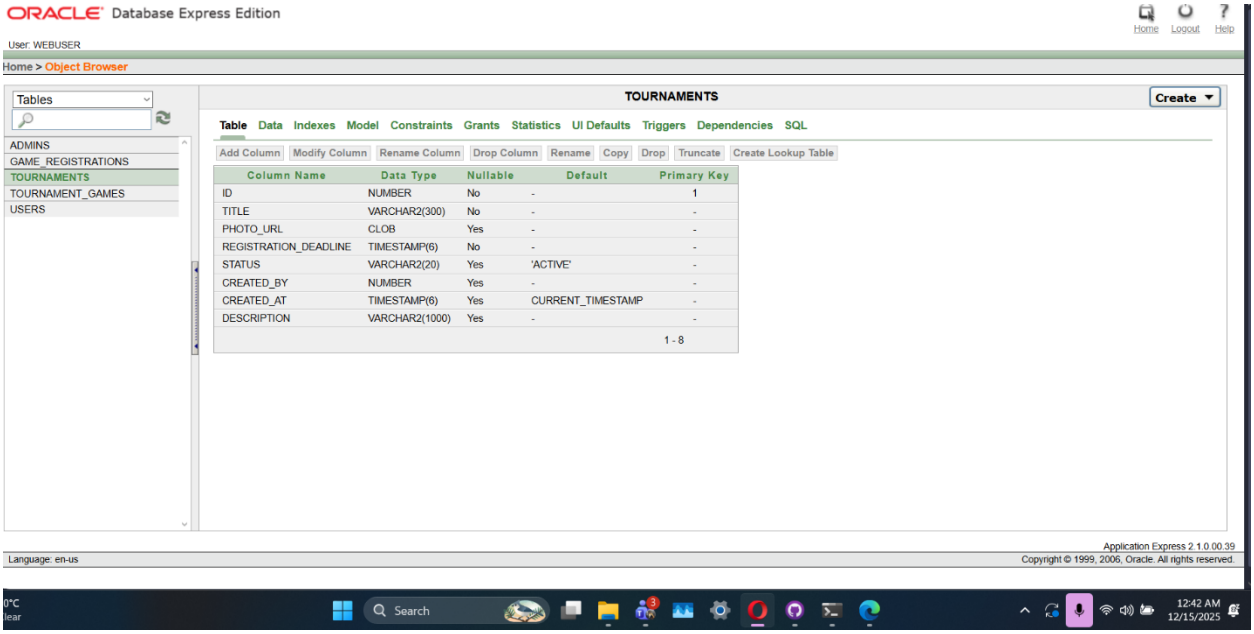
Admins Table

```
CREATE TABLE admins (
    id NUMBER PRIMARY KEY,
    admin_id VARCHAR2(50) NOT NULL UNIQUE,
    email VARCHAR2(100) NOT NULL UNIQUE,
    full_name VARCHAR2(200),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```



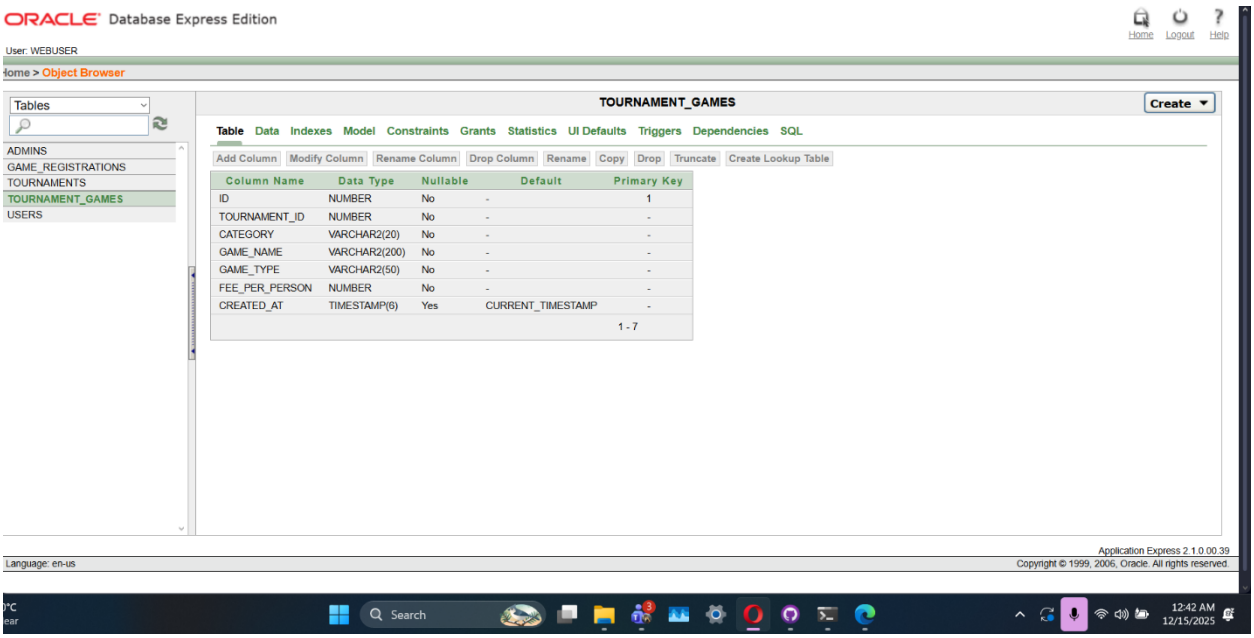
Tournaments Table

```
CREATE TABLE tournaments (
    id NUMBER PRIMARY KEY,
    title VARCHAR2(300) NOT NULL,
    photo_url CLOB,
    registration_deadline TIMESTAMP NOT NULL,
    status VARCHAR2(20) DEFAULT 'ACTIVE',
    created_by NUMBER REFERENCES admins(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_tournament_status CHECK (status IN ('ACTIVE', 'CLOSED', 'COMPLETED'))
);
```



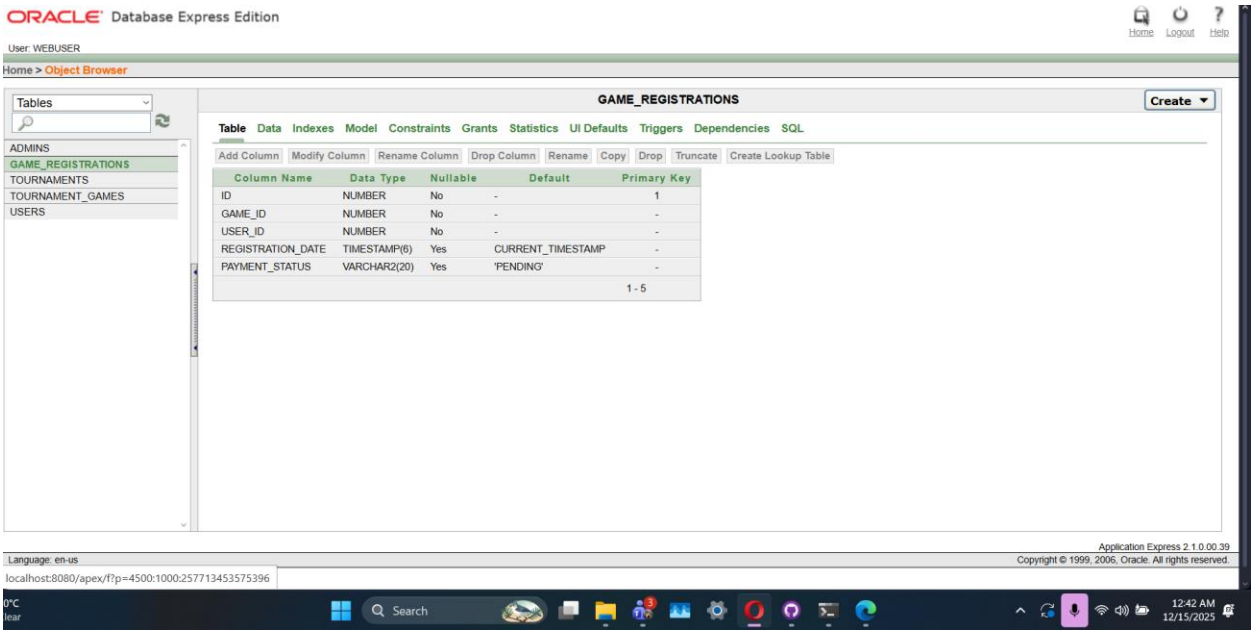
Tournament Games Table

```
CREATE TABLE tournament_games (
    id NUMBER PRIMARY KEY,
    tournament_id NUMBER NOT NULL REFERENCES tournaments(id) ON DELETE CASCADE,
    category VARCHAR2(20) NOT NULL,
    game_name VARCHAR2(200) NOT NULL,
    game_type VARCHAR2(50) NOT NULL,
    fee_per_person NUMBER NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_game_category CHECK (category IN ('Male', 'Female', 'Mix')),
    CONSTRAINT chk_game_type CHECK (game_type IN ('Solo', 'Duo', 'Custom'))
);
```



Game Registrations Table

```
CREATE TABLE game_registrations (
    id NUMBER PRIMARY KEY,
    game_id NUMBER NOT NULL REFERENCES tournament_games(id) ON DELETE CASCADE,
    user_id NUMBER NOT NULL REFERENCES users(id),
    registration_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    payment_status VARCHAR2(20) DEFAULT 'PENDING',
    CONSTRAINT chk_payment_status CHECK (payment_status IN ('PENDING', 'PAID', 'FAILED'))
);
```



Data Insertion Using SQL

Insert Data into Users table

INSERT INTO users (id, student_id, email, full_name, gender, phone_number, program_level, department, blood_group, profile_completed, name_edit_count, is_first_login)

VALUES (user_id_seq.NEXTVAL, '24-56434-1', '24-56434-1@student.aiub.edu', 'John Smith', 'Male', '01712345678', 'Undergraduate', 'CSE', 'A+', 1, 0, 0);

INSERT INTO users (id, student_id, email, full_name, gender, phone_number, program_level, department, blood_group, profile_completed, name_edit_count, is_first_login)

VALUES (user_id_seq.NEXTVAL, '23-45123-2', '23-45123-2@student.aiub.edu', 'Sarah Johnson', 'Female', '01823456789', 'Undergraduate', 'EEE', 'B+', 1, 1, 0);

INSERT INTO users (id, student_id, email, full_name, gender, phone_number, program_level, department, blood_group, profile_completed, name_edit_count, is_first_login)

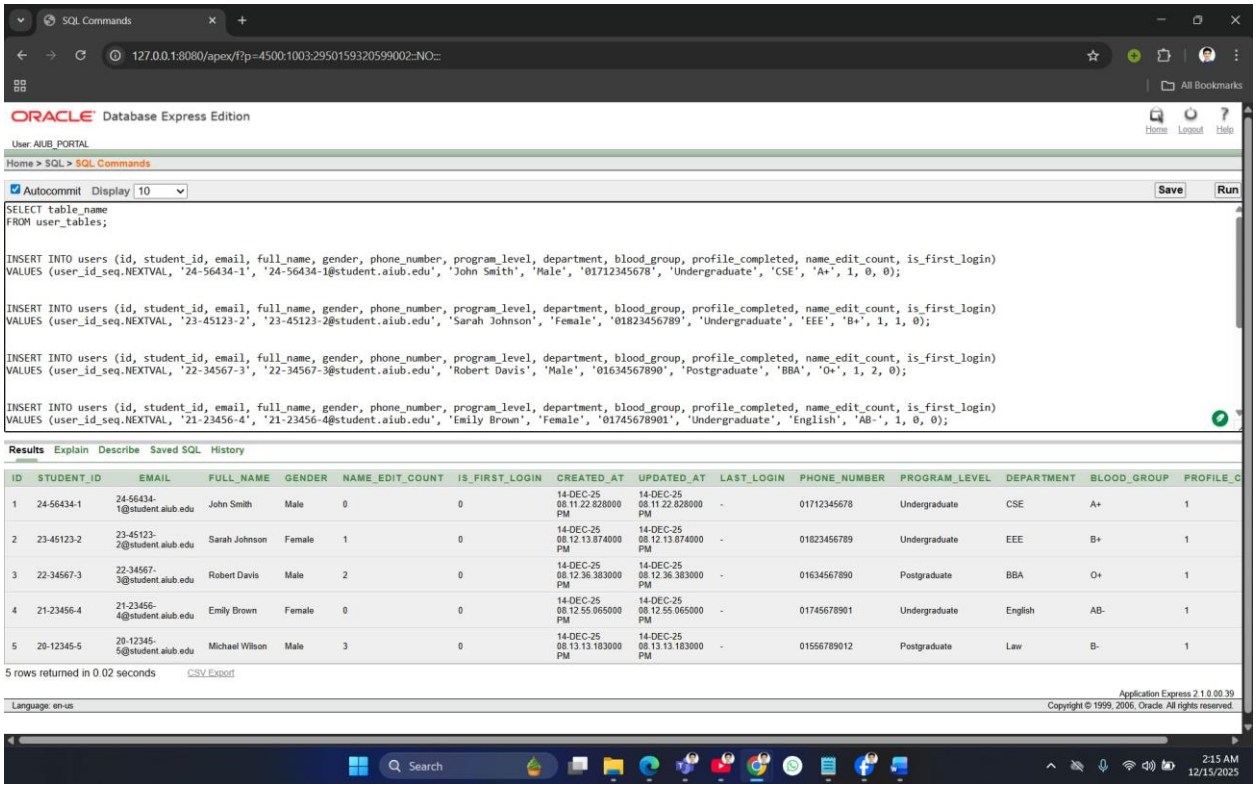
VALUES (user_id_seq.NEXTVAL, '22-34567-3', '22-34567-3@student.aiub.edu', 'Robert Davis', 'Male', '01634567890', 'Postgraduate', 'BBA', 'O+', 1, 2, 0);

INSERT INTO users (id, student_id, email, full_name, gender, phone_number, program_level, department, blood_group, profile_completed, name_edit_count, is_first_login)

VALUES (user_id_seq.NEXTVAL, '21-23456-4', '21-23456-4@student.aiub.edu', 'Emily Brown', 'Female', '01745678901', 'Undergraduate', 'English', 'AB-', 1, 0, 0);

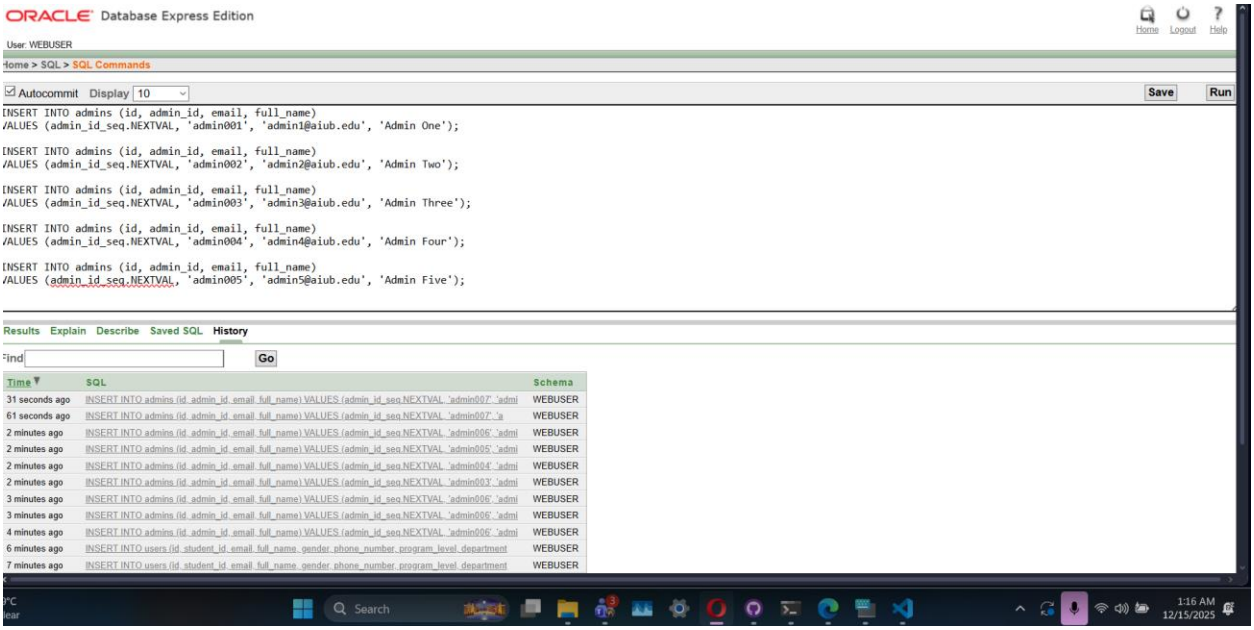
INSERT INTO users (id, student_id, email, full_name, gender, phone_number, program_level, department, blood_group, profile_completed, name_edit_count, is_first_login)

VALUES (user_id_seq.NEXTVAL, '20-12345-5', '20-12345-5@student.aiub.edu', 'Michael Wilson', 'Male', '01556789012', 'Postgraduate', 'Law', 'B-', 1, 3, 0);



Insert Data into Admin table

```
INSERT INTO admins (id, admin_id, email, full_name)
VALUES (admin_id_seq.NEXTVAL, 'admin001', 'admin1@aiub.edu', 'Admin One');
INSERT INTO admins (id, admin_id, email, full_name)
VALUES (admin_id_seq.NEXTVAL, 'admin002', 'admin2@aiub.edu', 'Admin Two');
INSERT INTO admins (id, admin_id, email, full_name)
VALUES (admin_id_seq.NEXTVAL, 'admin003', 'admin3@aiub.edu', 'Admin Three');
INSERT INTO admins (id, admin_id, email, full_name)
VALUES (admin_id_seq.NEXTVAL, 'admin004', 'admin4@aiub.edu', 'Admin Four');
INSERT INTO admins (id, admin_id, email, full_name)
VALUES (admin_id_seq.NEXTVAL, 'admin005', 'admin5@aiub.edu', 'Admin Five');
```



Insert Data into Tournaments table

INSERT INTO tournaments (id, title, photo_url, registration_deadline, status, created_by)

VALUES (tournament_id_seq.NEXTVAL, 'AIUB Annual Sports Festival 2025', 'https://example.com/images/sports_fest_2025.jpg', TO_DATE('2025-02-15 23:59:59', 'YYYY-MM-DD HH24:MI:SS'), 'ACTIVE', 1);

INSERT INTO tournaments (id, title, photo_url, registration_deadline, status, created_by)

VALUES (tournament_id_seq.NEXTVAL, 'Inter-Department Cricket Championship', 'https://example.com/images/cricket_champ.jpg', TO_DATE('2025-01-20 23:59:59', 'YYYY-MM-DD HH24:MI:SS'), 'ACTIVE', 2);

INSERT INTO tournaments (id, title, photo_url, registration_deadline, status, created_by)

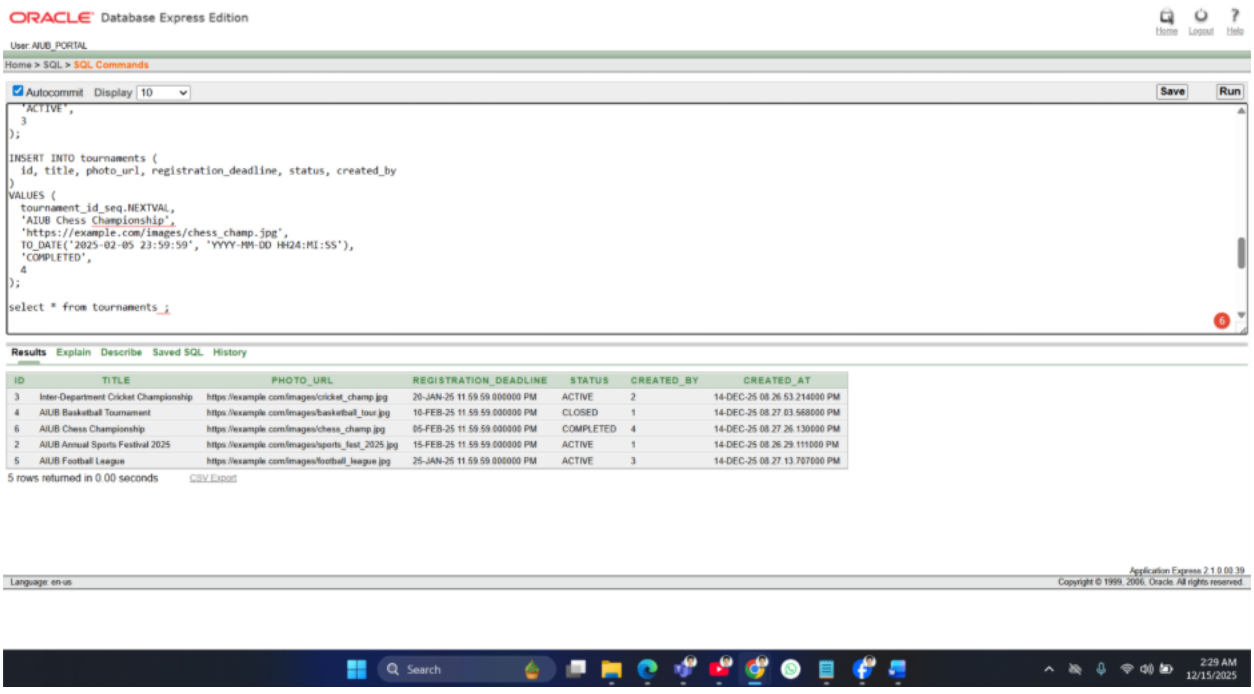
VALUES (tournament_id_seq.NEXTVAL, 'AIUB Basketball Tournament', 'https://example.com/images/basketball_tour.jpg', TO_DATE('2025-02-10 23:59:59', 'YYYY-MM-DD HH24:MI:SS'), 'CLOSED', 1);

INSERT INTO tournaments (id, title, photo_url, registration_deadline, status, created_by)

VALUES (tournament_id_seq.NEXTVAL, 'AIUB Football League', 'https://example.com/images/football_league.jpg', TO_DATE('2025-01-25 23:59:59', 'YYYY-MM-DD HH24:MI:SS'), 'ACTIVE', 3);

INSERT INTO tournaments (id, title, photo_url, registration_deadline, status, created_by)

VALUES (tournament_id_seq.NEXTVAL, 'AIUB Chess Championship', 'https://example.com/images/chess_champ.jpg', TO_DATE('2025-02-05 23:59:59', 'YYYY-MM-DD HH24:MI:SS'), 'COMPLETED', 4);



Insert Data into Tournament_games table

INSERT INTO tournament_games (id, tournament_id, category, game_name, game_type, fee_per_person)

VALUES (game_id_seq.NEXTVAL, 1, 'Male', 'Cricket', 'Custom', 500);

INSERT INTO tournament_games (id, tournament_id, category, game_name, game_type, fee_per_person)

VALUES (game_id_seq.NEXTVAL, 1, 'Female', 'Badminton', 'Solo', 300);

INSERT INTO tournament_games (id, tournament_id, category, game_name, game_type, fee_per_person)

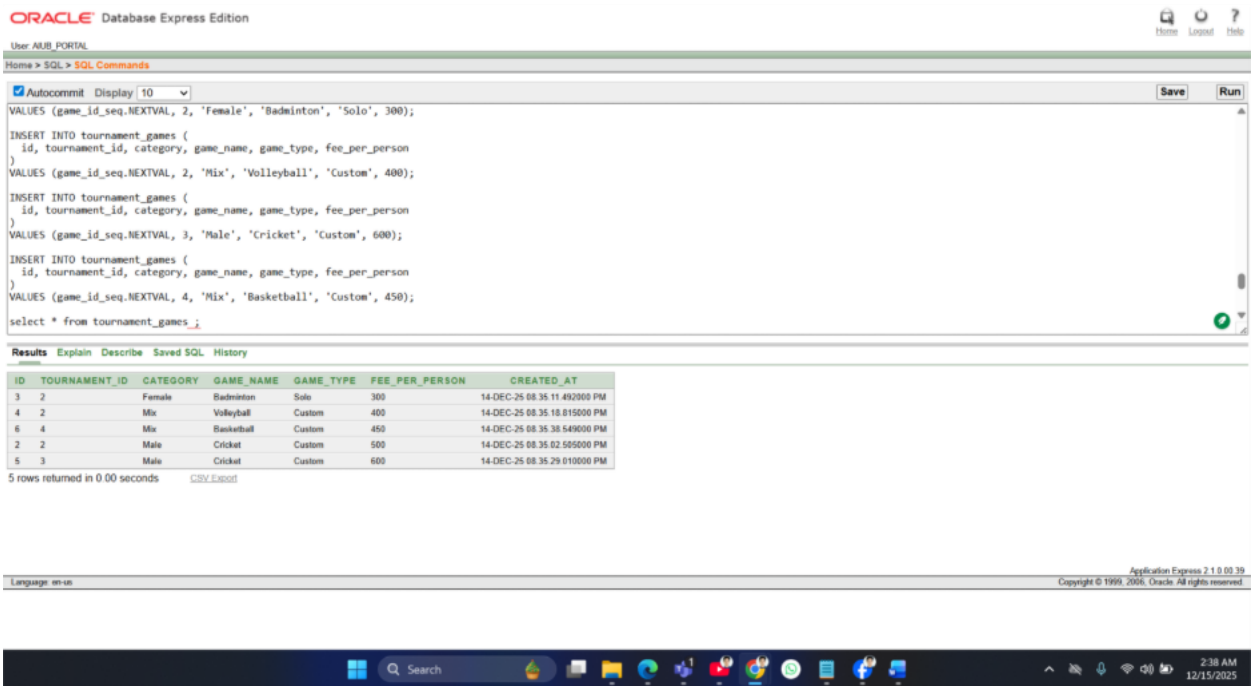
VALUES (game_id_seq.NEXTVAL, 1, 'Mix', 'Volleyball', 'Custom', 400);

INSERT INTO tournament_games (id, tournament_id, category, game_name, game_type, fee_per_person)

VALUES (game_id_seq.NEXTVAL, 2, 'Male', 'Cricket', 'Custom', 600);

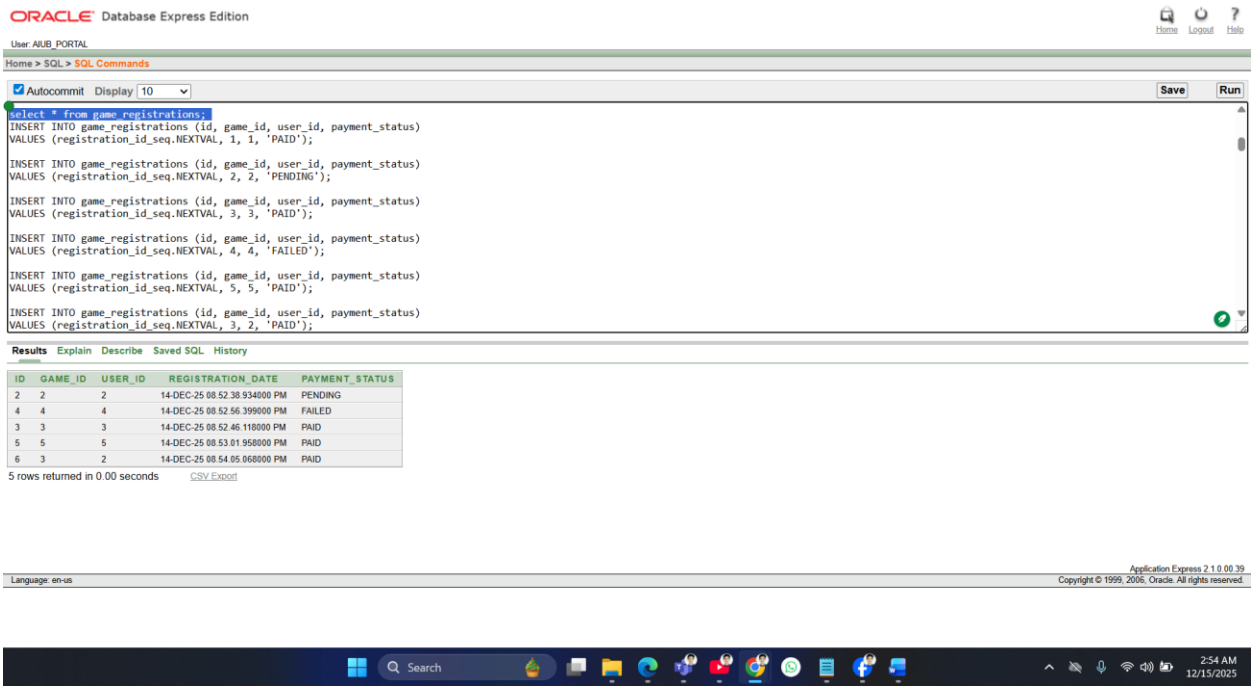
INSERT INTO tournament_games (id, tournament_id, category, game_name, game_type, fee_per_person)

VALUES (game_id_seq.NEXTVAL, 3, 'Mix', 'Basketball', 'Custom', 450);



Insert Data into Game_registrations table

```
INSERT INTO game_registrations (id, game_id, user_id, payment_status)
VALUES (registration_id_seq.NEXTVAL, 1, 1, 'PAID');
INSERT INTO game_registrations (id, game_id, user_id, payment_status)
VALUES (registration_id_seq.NEXTVAL, 2, 2, 'PENDING');
INSERT INTO game_registrations (id, game_id, user_id, payment_status)
VALUES (registration_id_seq.NEXTVAL, 3, 3, 'PAID');
INSERT INTO game_registrations (id, game_id, user_id, payment_status)
VALUES (registration_id_seq.NEXTVAL, 4, 4, 'FAILED');
INSERT INTO game_registrations (id, game_id, user_id, payment_status)
VALUES (registration_id_seq.NEXTVAL, 5, 5, 'PAID');
COMMIT;
```



Basic PL/SQL

2 Variables

1. DECLARE

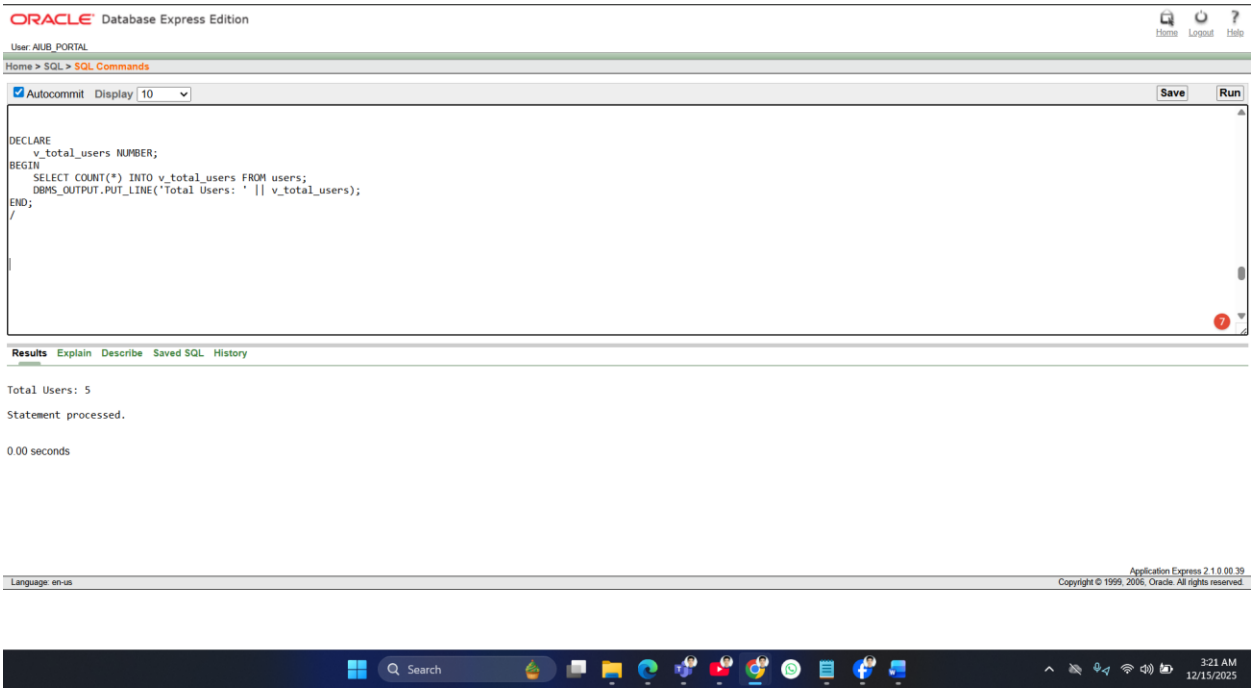
    v_total_users NUMBER;

BEGIN

    SELECT COUNT(*) INTO v_total_users FROM users;

    DBMS_OUTPUT.PUT_LINE('Total Users: ' || v_total_users);

END;



2. DECLARE

    v_first_user_name VARCHAR2(100);

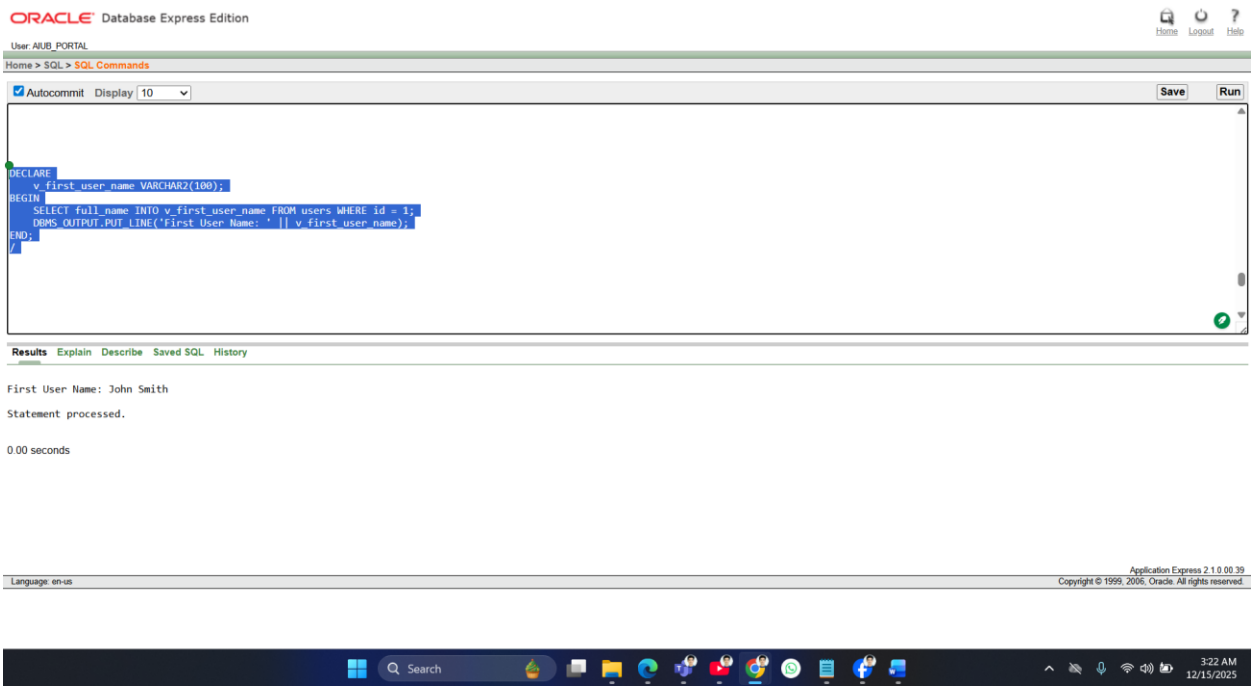
BEGIN

    SELECT full_name INTO v_first_user_name FROM users WHERE id = 1;

    DBMS_OUTPUT.PUT_LINE('First User Name: ' || v_first_user_name);

END;

/



2 Operators

1.DECLARE

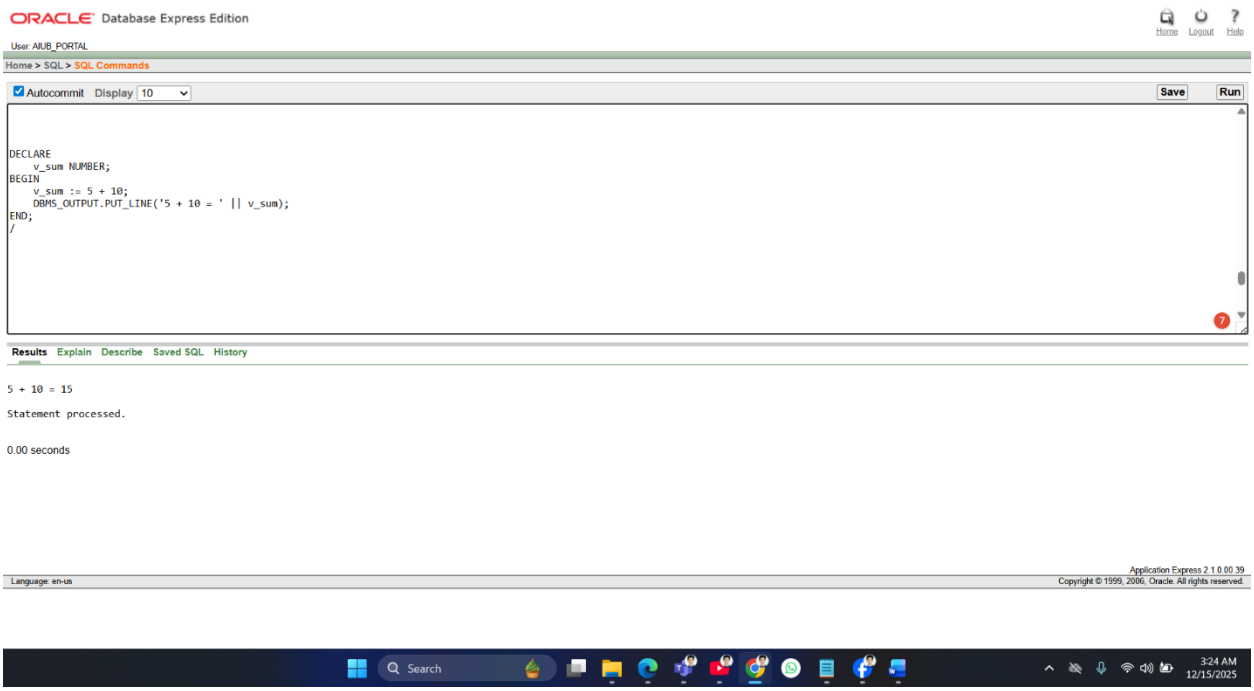
v_sum NUMBER;

BEGIN

v_sum := 5 + 10;

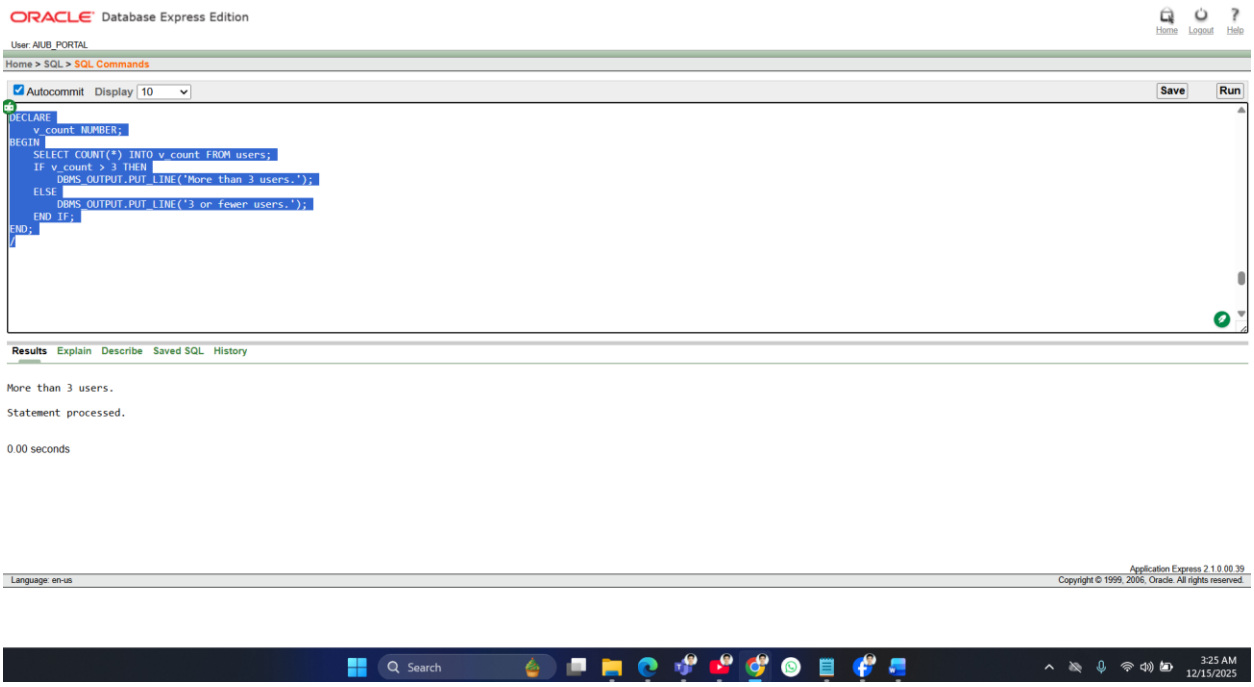
DBMS_OUTPUT.PUT_LINE('5 + 10 = ' || v_sum);

END;



2.DECLARE

```
   v_count NUMBER;
BEGIN
   SELECT COUNT(*) INTO v_count FROM users;
   IF v_count > 3 THEN
      DBMS_OUTPUT.PUT_LINE('More than 3 users.');
   ELSE
      DBMS_OUTPUT.PUT_LINE('3 or fewer users.');
   END IF;
END;
```



2 Single-row function

1. DECLARE

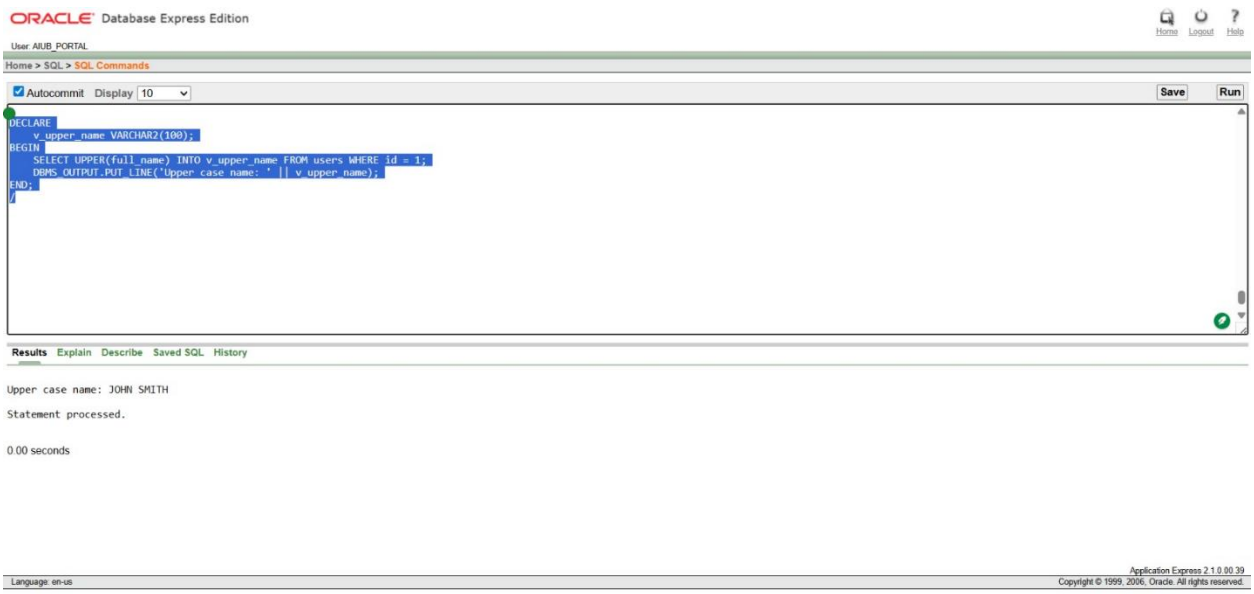
v_upper_name VARCHAR2(100);

BEGIN

SELECT UPPER(full_name) INTO v_upper_name FROM users WHERE id = 1;

DBMS_OUTPUT.PUT_LINE('Upper case name: ' || v_upper_name);

END;



2. DECLARE

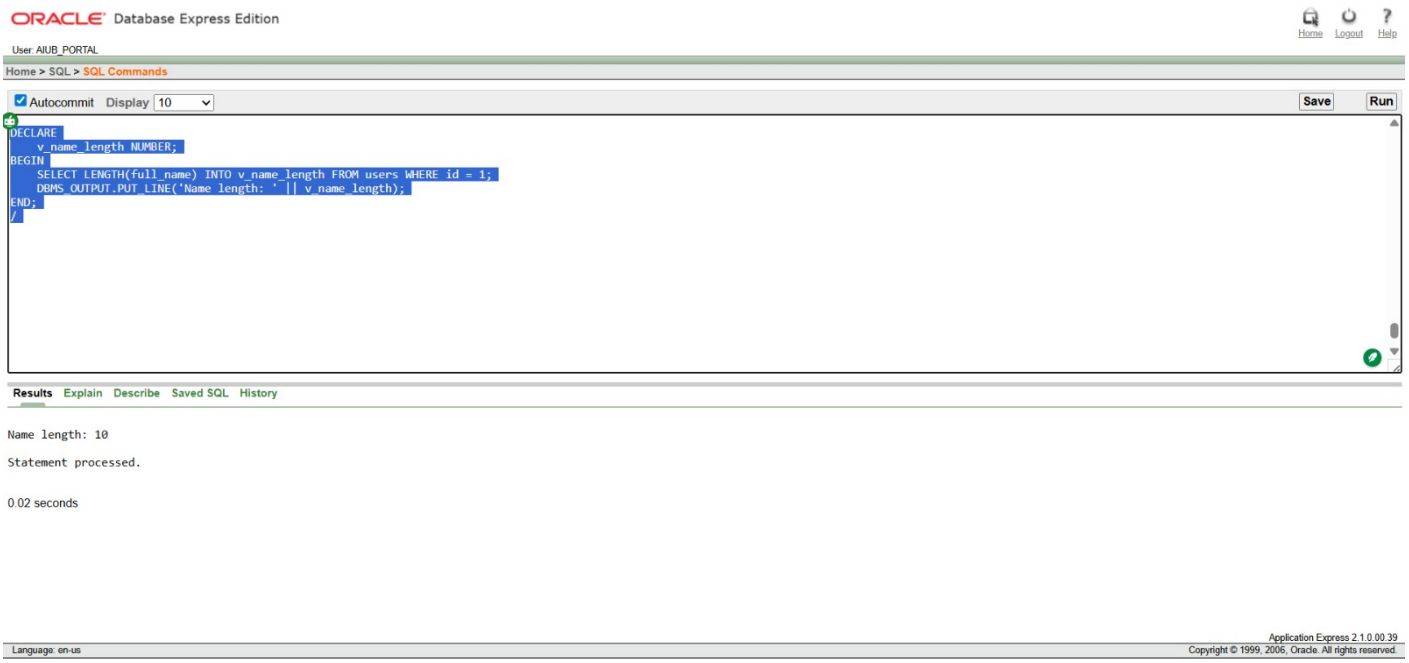
v_name_length NUMBER;

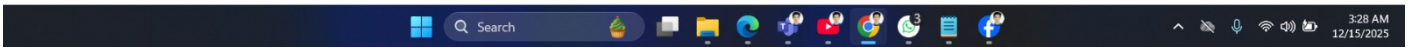
BEGIN

SELECT LENGTH(full_name) INTO v_name_length FROM users WHERE id = 1;

DBMS_OUTPUT.PUT_LINE('Name length: ' || v_name_length);

END;





2 group function

1. DECLARE

v_count NUMBER;

BEGIN

SELECT COUNT(*) INTO v_count FROM users;

DBMS_OUTPUT.PUT_LINE('Total Users: ' || v_count);

END;

2. DECLARE

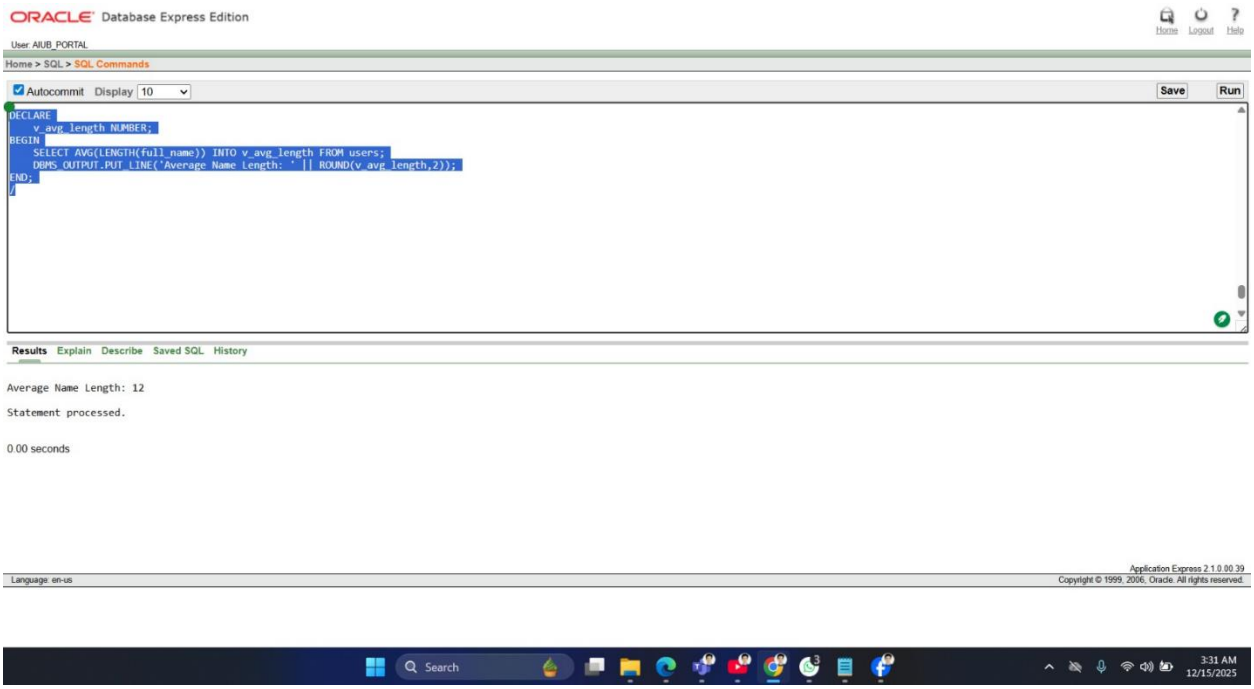
v_avg_length NUMBER;

BEGIN

SELECT AVG(LENGTH(full_name)) INTO v_avg_length FROM users;

DBMS_OUTPUT.PUT_LINE('Average Name Length: ' || ROUND(v_avg_length,2));

END;



2 conditional statements

1. DECLARE

v_count NUMBER;

BEGIN

SELECT COUNT(*) INTO v_count FROM users;

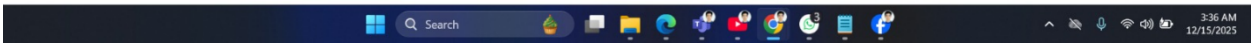
IF v_count > 3 THEN

DBMS_OUTPUT.PUT_LINE('More than 3 users.');

ELSE

DBMS_OUTPUT.PUT_LINE('3 or fewer users.');

END IF; END;



2.BEGIN

FOR rec IN (SELECT full_name, program_level FROM users) LOOP

CASE rec.program_level

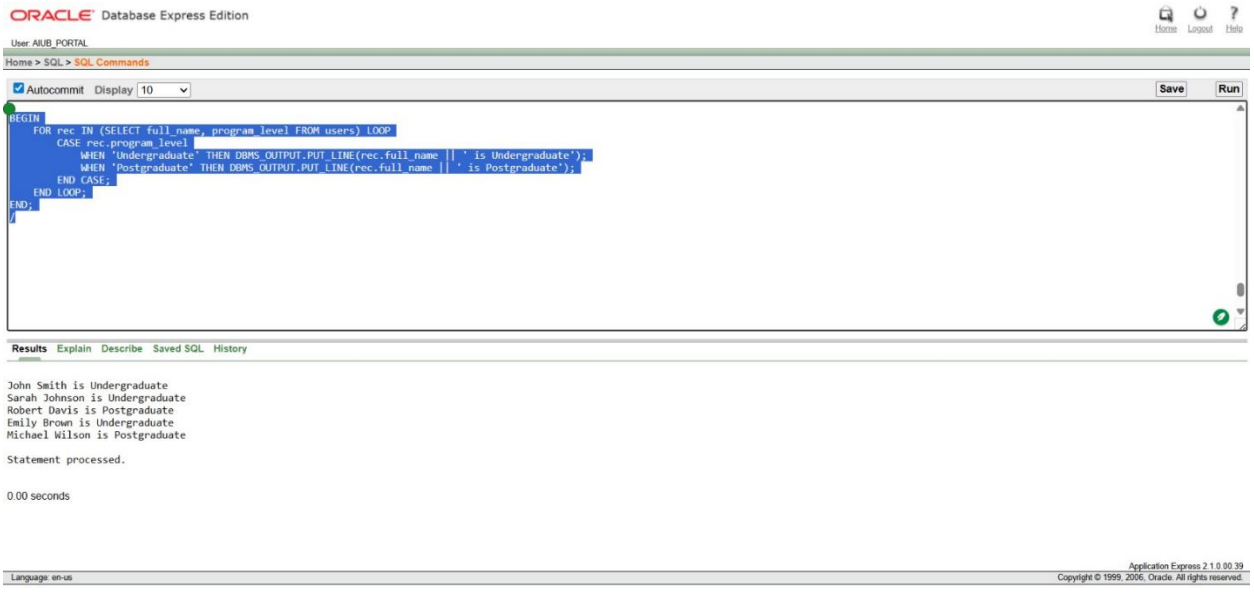
WHEN 'Undergraduate' THEN DBMS_OUTPUT.PUT_LINE(rec.full_name || ' is Undergraduate');

WHEN 'Postgraduate' THEN DBMS_OUTPUT.PUT_LINE(rec.full_name || ' is Postgraduate');

END CASE;

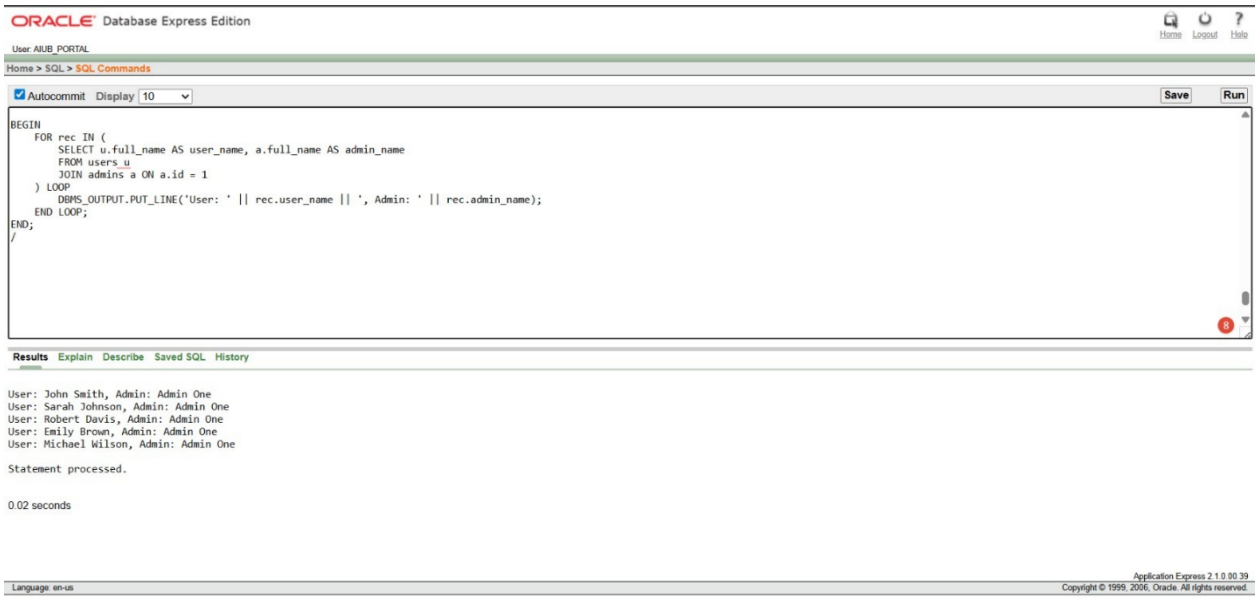
END LOOP;

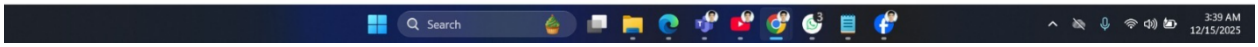
END;



2 Subquery

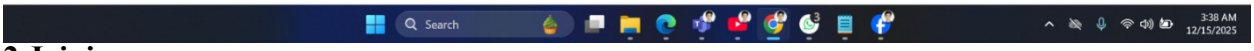
```
DECLARE
   v_dept_count NUMBER;
BEGIN
   SELECT COUNT(*) INTO v_dept_count
   FROM users
   WHERE department = (SELECT department FROM users WHERE id = 1);
   DBMS_OUTPUT.PUT_LINE('Users in same department as user 1: ' || v_dept_count);
END;
```





2.

```
DECLARE
   v_email VARCHAR2(100);
BEGIN
   SELECT email INTO v_email
   FROM users
   WHERE id = (SELECT MAX(id) FROM users);
   DBMS_OUTPUT.PUT_LINE('Email of user with max ID: ' || v_email);
END;
```



2 Joining

1. BEGIN

FOR rec IN (

SELECT u.full_name AS user_name, a.full_name AS admin_name

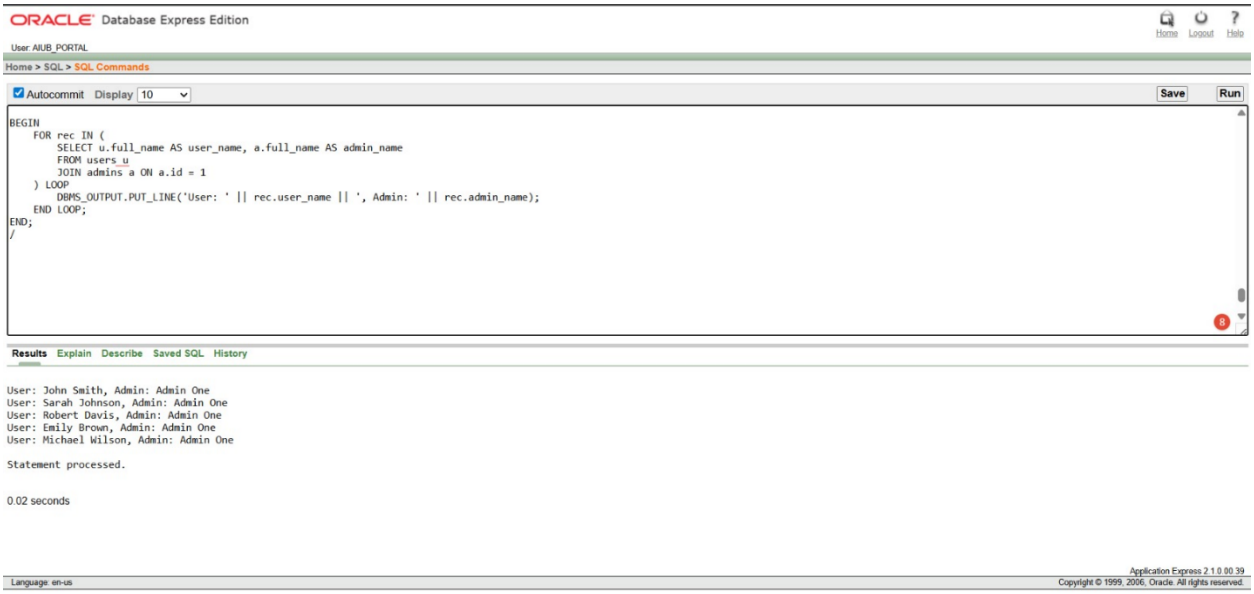
FROM users u

JOIN admins a ON a.id = 1) LOOP

DBMS_OUTPUT.PUT_LINE('User: ' || rec.user_name || ', Admin: ' || rec.admin_name);

END LOOP;

END;



2. BEGIN

FOR rec IN (

SELECT u.full_name AS user_name, a.full_name AS admin_name

FROM users u, admins a

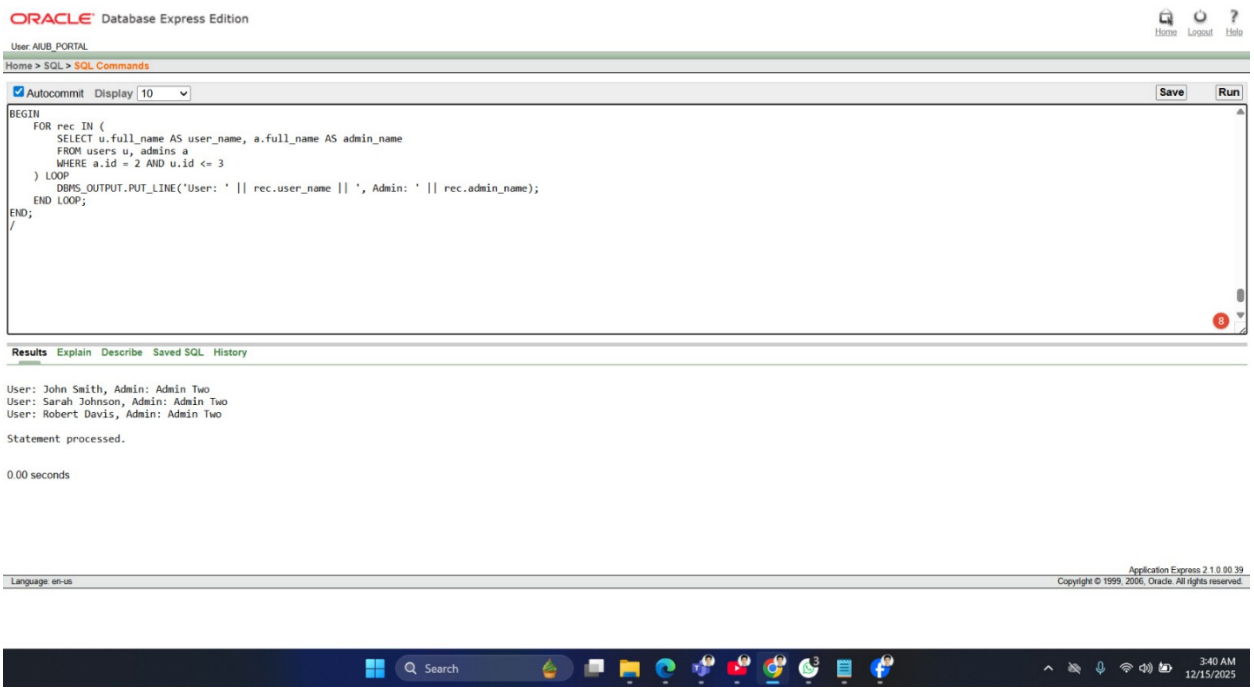
WHERE a.id = 2 AND u.id <= 3

) LOOP

DBMS_OUTPUT.PUT_LINE('User: ' || rec.user_name || ', Admin: ' || rec.admin_name);

END LOOP;

END;



Advance PL/SQL

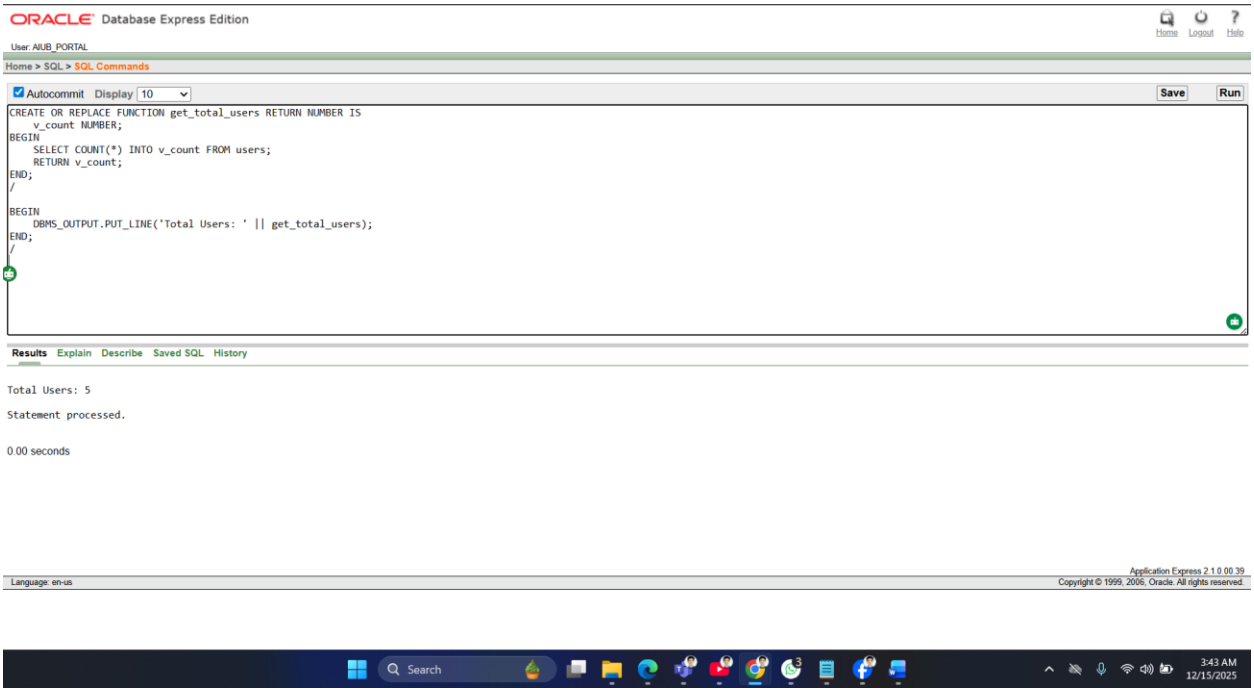
2 Stored functions

Function 1: Get total users

```
CREATE OR REPLACE FUNCTION get_total_users RETURN NUMBER IS
    v_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_count FROM users;
    RETURN v_count;
END;
/
```

Test Function 1

```
BEGIN
    DBMS_OUTPUT.PUT_LINE('Total Users: ' || get_total_users);
END;
/
```

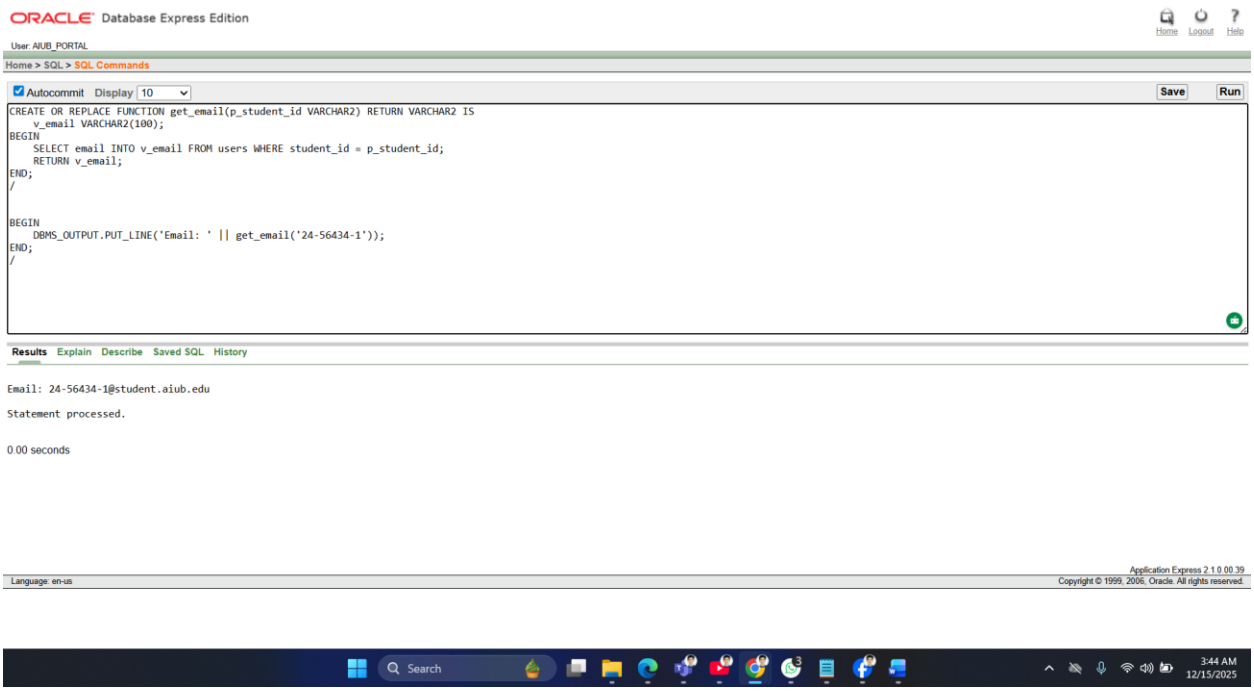


Function 2: Get email by student_id

```
CREATE OR REPLACE FUNCTION get_email(p_student_id VARCHAR2) RETURN
VARCHAR2 IS
    v_email VARCHAR2(100);
BEGIN
    SELECT email INTO v_email FROM users WHERE student_id = p_student_id;
    RETURN v_email;
END;
/
```

Test Function 2

```
BEGIN
    DBMS_OUTPUT.PUT_LINE('Email: ' || get_email('24-56434-1'));
END;
/
```

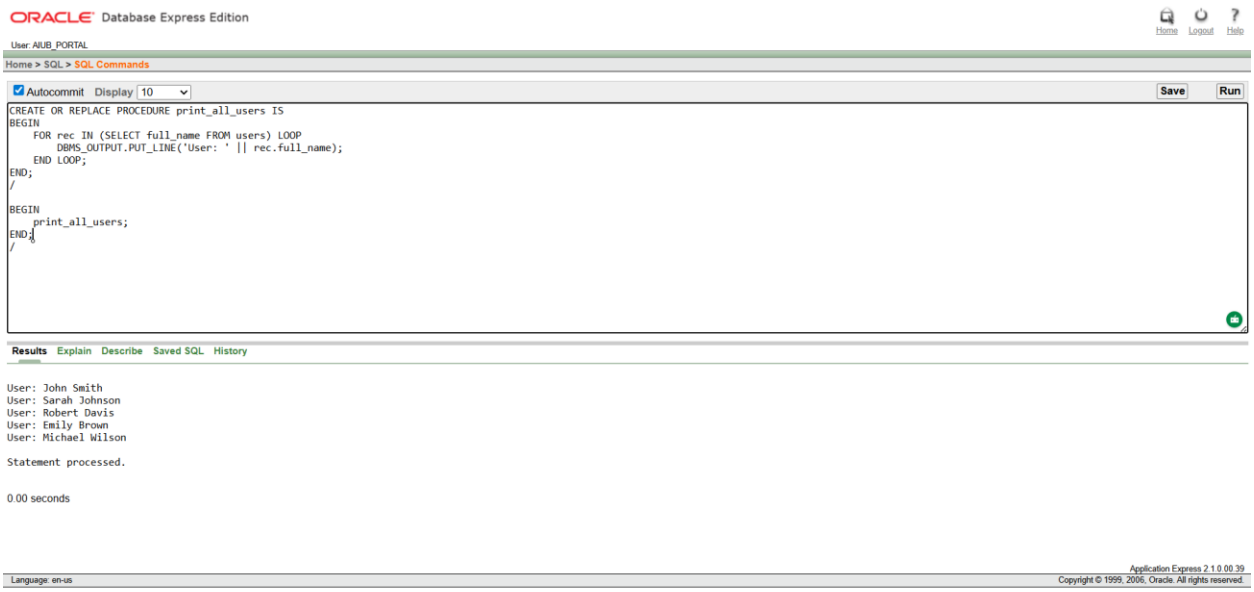


2. Stored Procedures

Procedure 1: Print all usernames

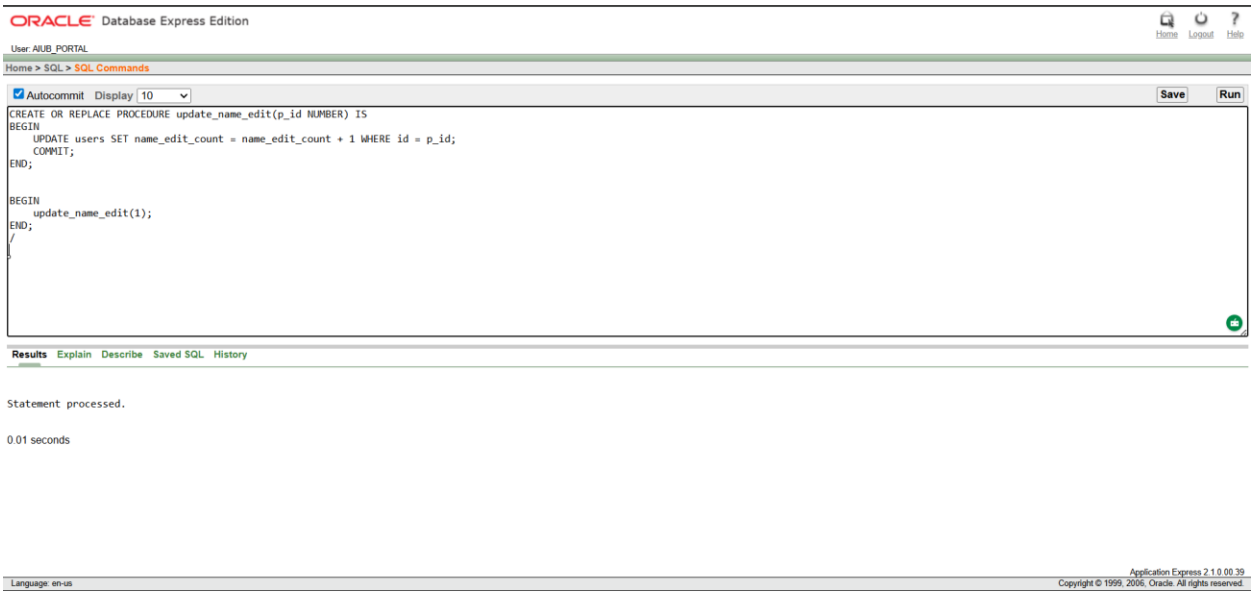
```
CREATE OR REPLACE PROCEDURE print_all_users IS
BEGIN
    FOR rec IN (SELECT full_name FROM users) LOOP
```

```
    DBMS_OUTPUT.PUT_LINE('User: ' || rec.full_name);
  END LOOP;
END;
/
-- Test Procedure 1
BEGIN
  print_all_users;
END;
/
```



Procedure 2: Update name_edit_count for a student

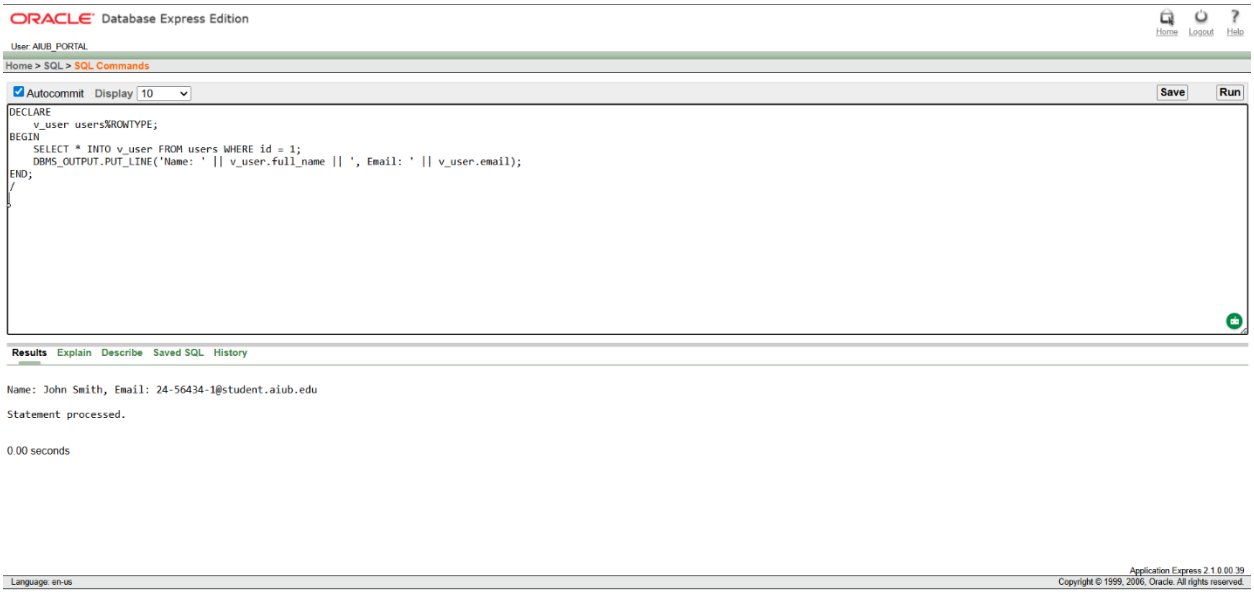
```
CREATE OR REPLACE PROCEDURE update_name_edit(p_id NUMBER) IS
BEGIN
  UPDATE users SET name_edit_count = name_edit_count + 1 WHERE id = p_id;
  COMMIT;
END;
Test Procedure 2
BEGIN
  update_name_edit(1); END;
```



2 Table-Based Record

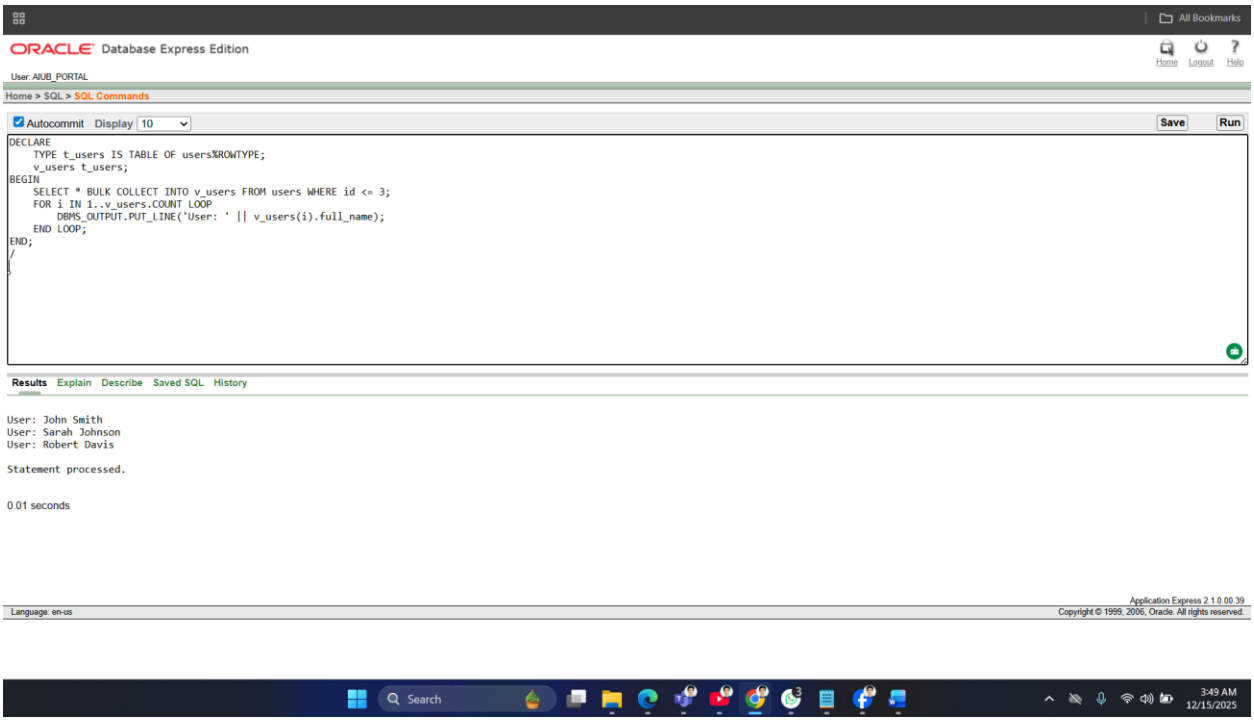
Example 1: Using %ROWTYPE

```
DECLARE
   v_user users%ROWTYPE;
BEGIN
   SELECT * INTO v_user FROM users WHERE id = 1;
   DBMS_OUTPUT.PUT_LINE('Name: ' || v_user.full_name || ', Email: ' || v_user.email);
END;
```



Example 2: Table of records

```
DECLARE
   TYPE t_users IS TABLE OF users%ROWTYPE;
   v_users t_users;
BEGIN
   SELECT * BULK COLLECT INTO v_users FROM users WHERE id <= 3;
   FOR i IN 1..v_users.COUNT LOOP
      DBMS_OUTPUT.PUT_LINE('User: ' || v_users(i).full_name);
   END LOOP;
END;
```

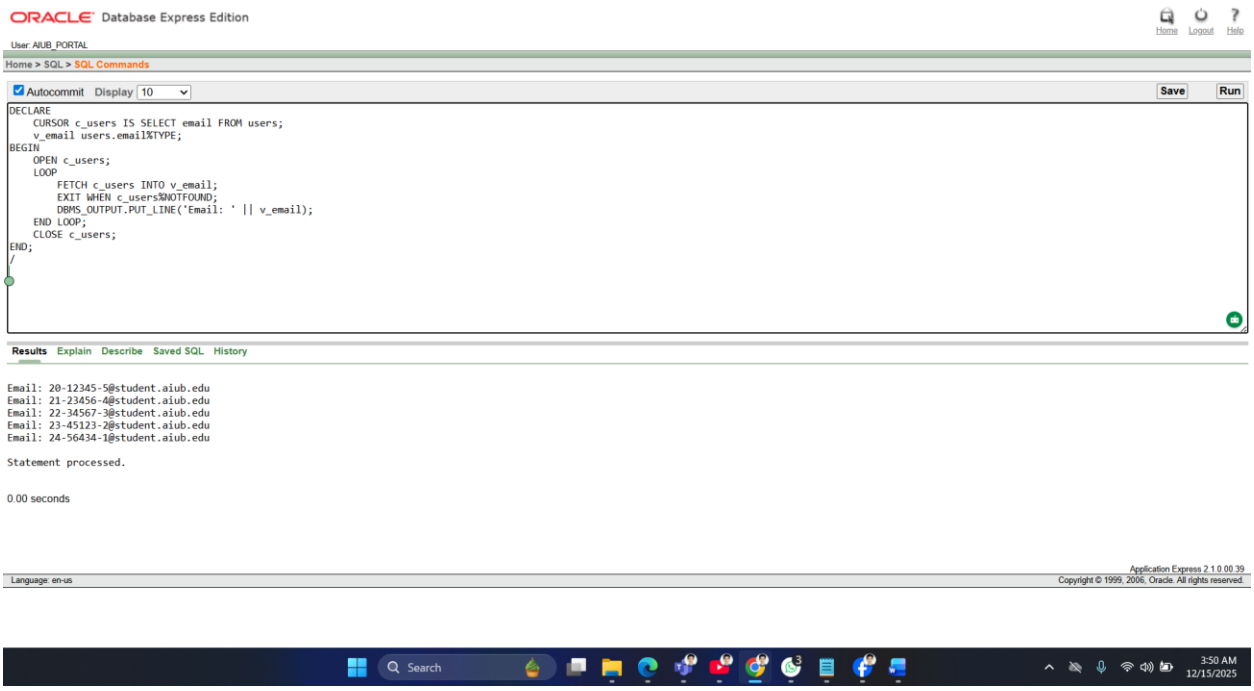


2 Explicit Cursor

Cursor 1: Print all emails

```
DECLARE
   CURSOR c_users IS SELECT email FROM users;
   v_email users.email%TYPE;
BEGIN
```

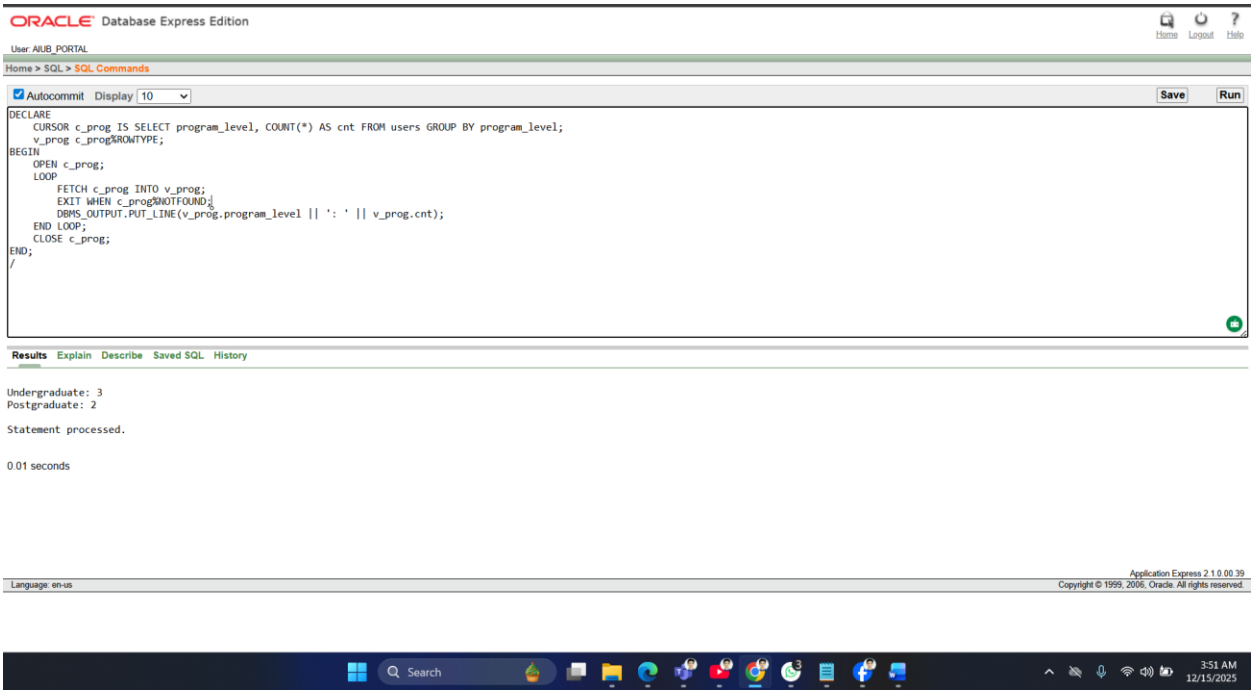
```
    OPEN c_users;
    LOOP
        FETCH c_users INTO v_email;
        EXIT WHEN c_users%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE('Email: ' || v_email);
    END LOOP;
    CLOSE c_users;
END;
/
```



Cursor 2: Count users by program_level

```
DECLARE
    CURSOR c_prog IS SELECT program_level, COUNT(*) AS cnt FROM users GROUP BY program_level;
    v_prog c_prog%ROWTYPE;
BEGIN
    OPEN c_prog;
    LOOP
```

```
    FETCH c_prog INTO v_prog;
    EXIT WHEN c_prog%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(v_prog.program_level || ': ' || v_prog.cnt);
  END LOOP;
  CLOSE c_prog;
END;
```

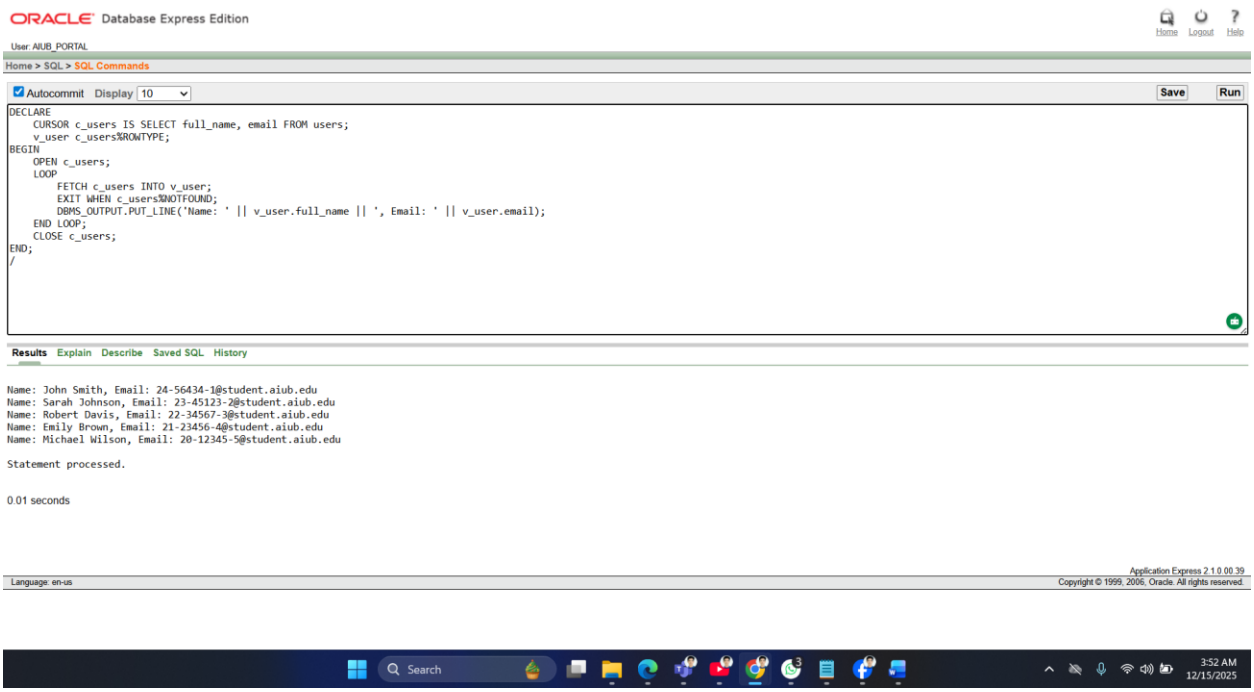


2 Cursor-Based Record

Example 1: Cursor with record

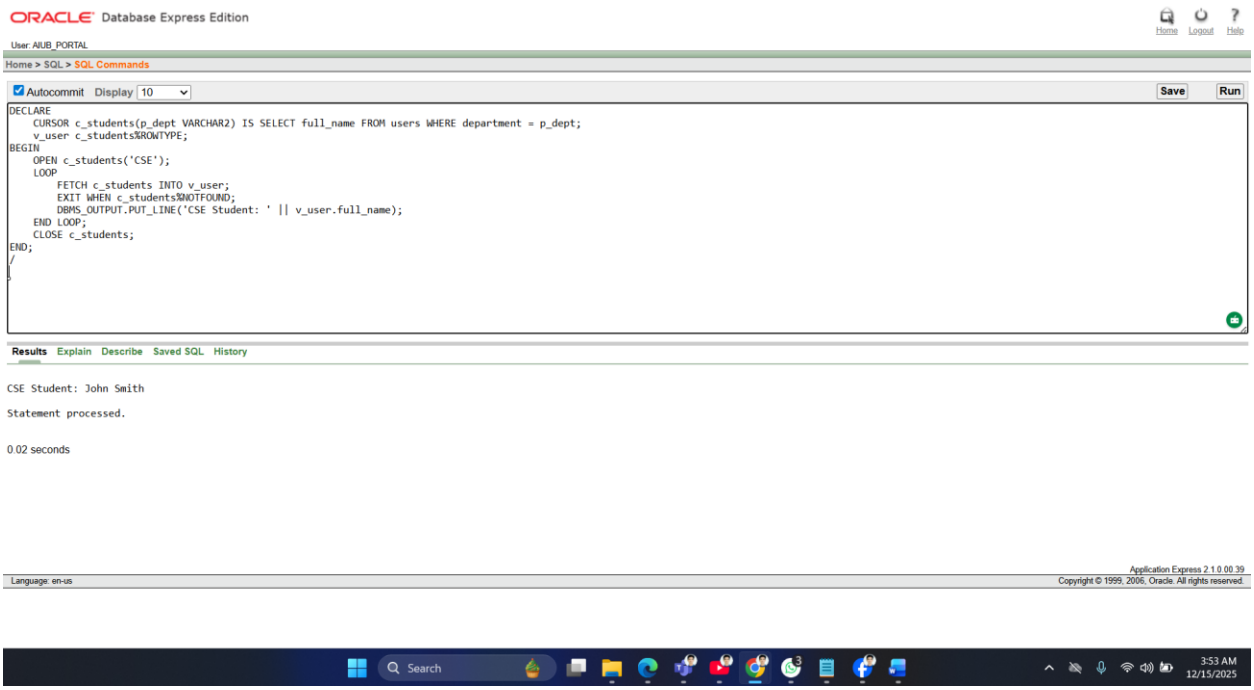
```
DECLARE
  CURSOR c_users IS SELECT full_name, email FROM users;
  v_user c_users%ROWTYPE;
BEGIN
  OPEN c_users;
  LOOP
    FETCH c_users INTO v_user;
    EXIT WHEN c_users%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE('Name: ' || v_user.full_name || ', Email: ' || v_user.email);
```

```
    END LOOP;
    CLOSE c_users;
END;
/
```



Example 2: Cursor with parameter

```
DECLARE
    CURSOR c_students(p_dept VARCHAR2) IS SELECT full_name FROM users WHERE department = p_dept;
    v_user c_students%ROWTYPE;
BEGIN
    OPEN c_students('CSE');
    LOOP
        FETCH c_students INTO v_user;
        EXIT WHEN c_students%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE('CSE Student: ' || v_user.full_name);
    END LOOP;
    CLOSE c_students;
END;
```

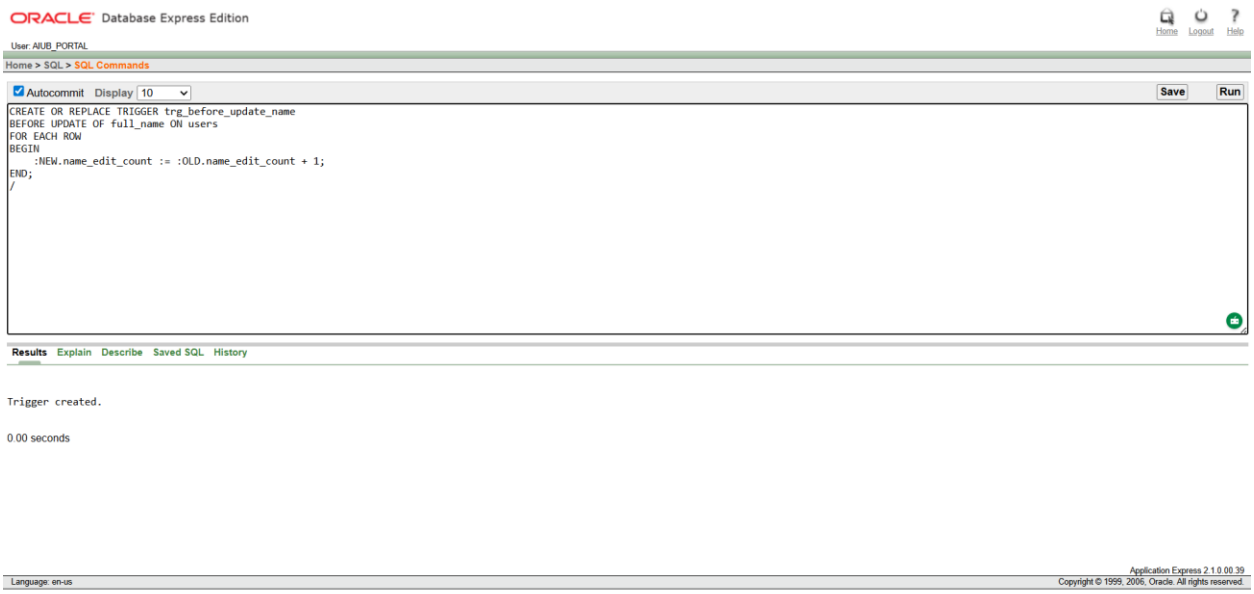


2 Row-Level Trigger

Trigger 1: Before insert, set profile_completed = 0

```
CREATE OR REPLACE TRIGGER trg_before_insert_users
BEFORE INSERT ON users
FOR EACH ROW
BEGIN
    :NEW.profile_completed := 0;
END;
```

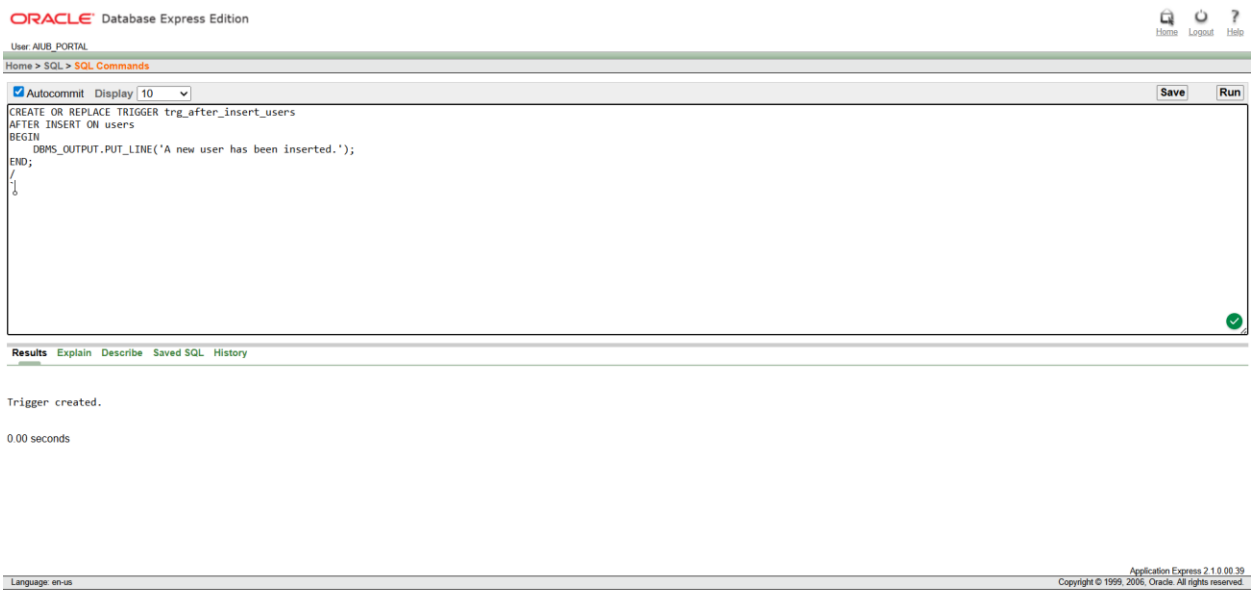
```
CREATE OR REPLACE TRIGGER trg_before_update_name
BEFORE UPDATE OF full_name ON users
FOR EACH ROW
BEGIN
    :NEW.name_edit_count := :OLD.name_edit_count + 1;
END;
```



2Statement-Level Trigger

Trigger 1: After insert on users, log message

```
CREATE OR REPLACE TRIGGER trg_after_insert_users
AFTER INSERT ON users
BEGIN
    DBMS_OUTPUT.PUT_LINE('A new user has been inserted.');
END;
```



Trigger 2: After delete on users, log message

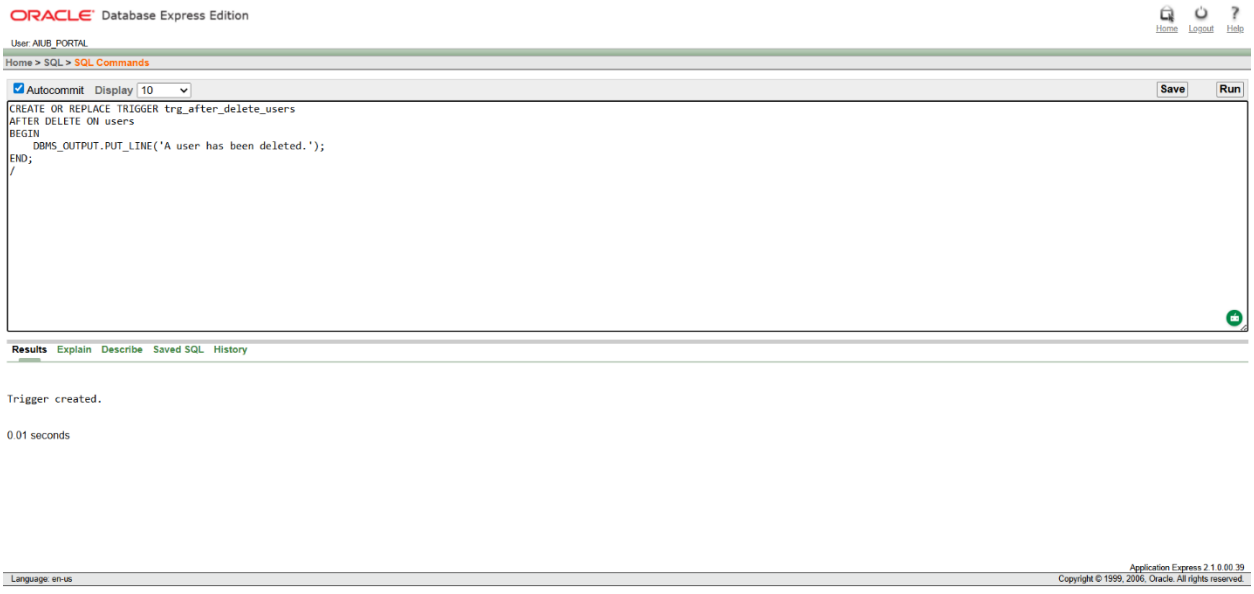
CREATE OR REPLACE TRIGGER trg_after_delete_users

AFTER DELETE ON users

BEGIN

DBMS_OUTPUT.PUT_LINE('A user has been deleted.');

END;



2 Package

1.CREATE OR REPLACE PACKAGE pkg_users IS

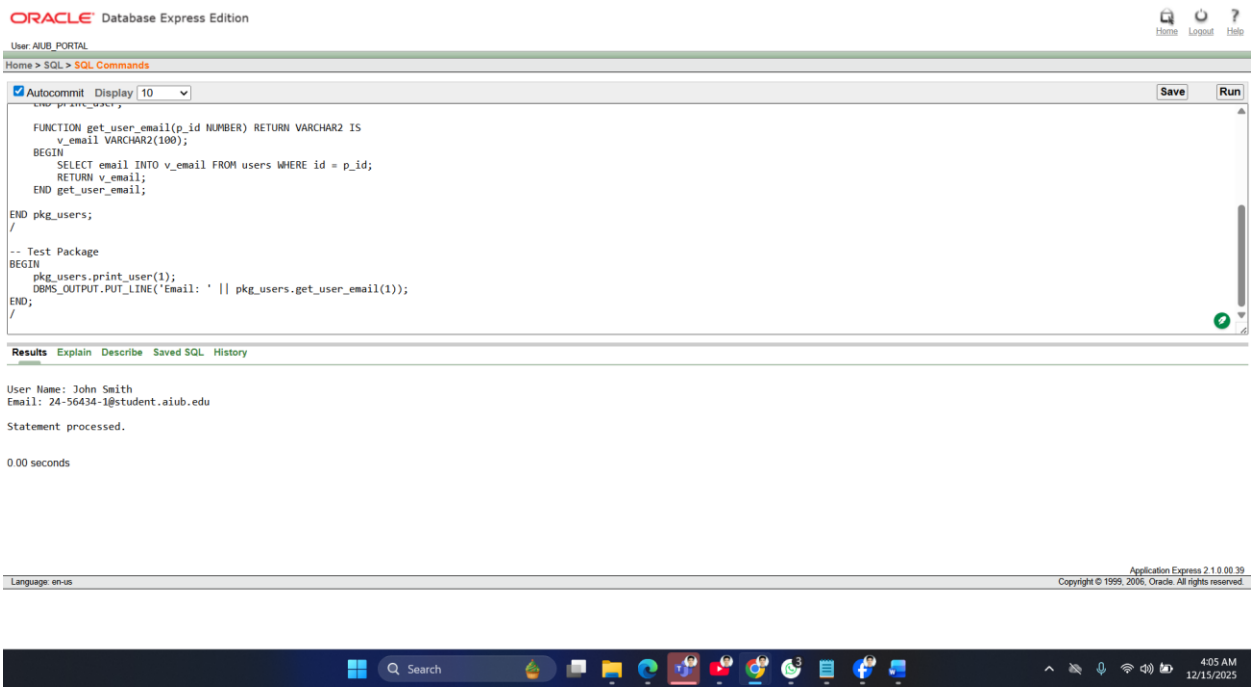
```
   PROCEDURE print_user(p_id NUMBER);
   FUNCTION get_user_email(p_id NUMBER) RETURN VARCHAR2;
END pkg_users;
/
```

Package Body

```
CREATE OR REPLACE PACKAGE BODY pkg_users IS

   PROCEDURE print_user(p_id NUMBER) IS
      v_name VARCHAR2(100);
   BEGIN
      SELECT full_name INTO v_name FROM users WHERE id = p_id;
      DBMS_OUTPUT.PUT_LINE('User Name: ' || v_name);
   END print_user;

   FUNCTION get_user_email(p_id NUMBER) RETURN VARCHAR2 IS
      v_email VARCHAR2(100);
   BEGIN
      SELECT email INTO v_email FROM users WHERE id = p_id;
      RETURN v_email;
   END get_user_email;

END pkg_users;
/
```

Test Package

```
BEGIN
   pkg_users.print_user(1);
   DBMS_OUTPUT.PUT_LINE('Email: ' || pkg_users.get_user_email(1));
END;
```



2. CREATE OR REPLACE PACKAGE pkg_users_info IS

PROCEDURE print_user_department(p_id NUMBER);

FUNCTION get_user_phone(p_id NUMBER) RETURN VARCHAR2;

END pkg_users_info;

/

CREATE OR REPLACE PACKAGE BODY pkg_users_info IS

PROCEDURE print_user_department(p_id NUMBER) IS

v_dept VARCHAR2(50);

BEGIN

SELECT department INTO v_dept FROM users WHERE id = p_id;

DBMS_OUTPUT.PUT_LINE('User Department: ' || v_dept);

END print_user_department;

FUNCTION get_user_phone(p_id NUMBER) RETURN VARCHAR2 IS

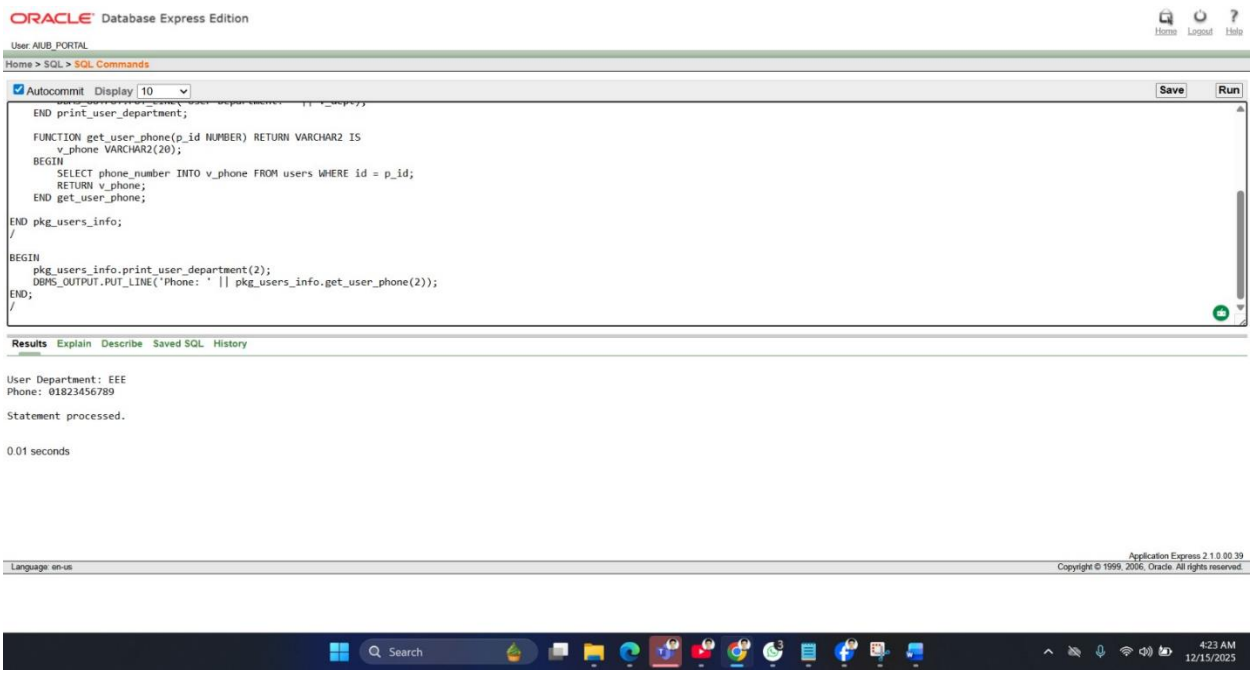
v_phone VARCHAR2(20);

BEGIN

SELECT phone_number INTO v_phone FROM users WHERE id = p_id;

RETURN v_phone;

```
    END get_user_phone;

END pkg_users_info;
/
BEGIN
    pkg_users_info.print_user_department(2);
    DBMS_OUTPUT.PUT_LINE('Phone: ' || pkg_users_info.get_user_phone(2));
END;
/
```



Conclusion

The AIUB Sports Portal was created to make sports management at American International University–Bangladesh easier and more organized. It successfully replaces the old manual registration system with a digital platform where students can log in securely, manage their profiles, view tournaments, and register for games online. At the same time, administrators can create tournaments, add games, and manage registrations from one place. This system reduces paperwork, saves time, and helps avoid common mistakes. The portal uses a well-structured database and secure login system to keep data accurate and safe. Overall, the project shows how a simple and well-designed web application can improve the management of university sports activities. In the future, the system can be improved by adding features such as online payment, email or SMS notifications, and match schedules. Live score updates, team management, and result tracking can also be included. Making the portal mobile-friendly or developing a mobile app would make it more convenient for users. With these improvements, the AIUB Sports Portal can become a more powerful and complete sports management system for AIUB.